Explicit block-encoding for partial differential equation-constrained optimization

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Partial differential equation (PDE)–constrained optimization, where an optimization problem is subject to PDE constraints, arises in various applications such as design, control, and inference. Solving such problems is computationally demanding because it requires repeatedly solving a PDE and using its solution within an optimization process. In this paper, we first propose a fully coherent quantum algorithm for solving PDE-constrained optimization problems. The proposed method combines a quantum PDE solver that prepares the solution vector as a quantum state, and a quantum optimizer that assumes oracle access to a quantized objective function. The central idea is the explicit construction of the oracle in a form of block-encoding for the objective function, which coherently uses the output of a quantum PDE solver. This enables us to avoid classical access to the full solution that requires quantum state tomography canceling out the potential quantum speedups. We also derive the overall computational complexity of the proposed method with respect to parameters for optimization and PDE simulation, where quantum speedup is inherited from the underlying quantum PDE solver. We numerically demonstrate the validity of the proposed method by applications, including a parameter calibration problem in the Black-Scholes equation and a material parameter design problem in the wave equation. This work presents the concept of composing quantum subroutines so that the weakness of one (i.e., prohibitive readout overhead) is neutralized by the strength of another (i.e., coherent oracle access), toward a bottleneck-free quantum algorithm.

I. INTRODUCTION

Partial differential equation (PDE)–constrained optimization [1, 2] arises in various applications, including design optimization [3, 4] such as shape [5, 6] and topology optimization [7], optimal control [8], and parameter estimation of a physical system. PDE-constrained optimization is a kind of inverse problems where one seeks design or control parameters ξ so that the state u of a physical system, governed by a PDE, minimizes an objective function. In an abstract setting, one solves the following problem:

$$\min_{\xi\in\Xi_{\mathrm{ad}}} \ \mathcal{F}(u,\xi) \text{ subject to } \ \mathcal{G}(u,\xi)=0, \tag{1}$$

where Ξ_{ad} is an admissible space of design parameters, $\mathcal{F}$ is an objective function, and $\mathcal{G}$ is a governing equation of a physical system of interest given by PDEs with appropriate boundary conditions. If we consider $\Xi_{\mathrm{ad}}\subset\mathbb{R}^M$ with M the number of parameters, we have continuous optimization, while if $\Xi_{\mathrm{ad}}\subset\mathbb{Z}^M$, we have discrete optimization. When focusing on continuous optimization, one typically relies on gradient-based approaches to solve the problem in Eq. (1), where each optimization step requires at least one forward and adjoint numerical analyses of PDE $\mathcal{G}$ to compute the objective function $\mathcal{F}$ and its derivative $\nabla_\xi\mathcal{F}$. For discrete optimization, one usually uses heuristics in which many forward numerical analyses are required to compute the objective function $\mathcal{F}$. As system sizes of PDEs grow, the computational cost of these numerical analyses can be computationally prohibitive, which makes it intractable to solve PDE-constrained optimization problems.

So, could quantum computing provide an efficient solution to those problems? In fact, first, quantum computing offers asymptotic speedups for linear-algebra primitives underlying the numerical analysis of PDEs. For stationary (elliptic) problems, spatial discretization yields a sparse linear system, which can be solved by quantum linear system algorithms (QLSAs) such as the HHL [9], linear combination of unitaries (LCU)-based [10] and quantum singular value transformation (QSVT)-based algorithms [11]. Under standard access models (sparse-access oracles for HHL and LCU; block-encoding for QSVT), these algorithms can offer quantum speedups with respect to the system size N of discretized PDEs. Several works proposed methods for solving elliptic PDEs by quantum algorithms [12, 13]. For evolution (parabolic or hyperbolic) problems, space and time discretization yield a linear system, which can be solved by QL-

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SAs [14]. Also, spatial discretization results in an ordinary differential equation (ODE), which can be simulated by quantum ODE solvers such as linear combination of Hamiltonian simulation (LCHS) [15, 16] and Schrödingerization [17, 18]. These approaches can provide quantum speedups in terms of the system size N of discretized PDEs under Hamiltonian simulation oracles, serving as an access model. There are several works that applied these algorithms or simply Hamiltonian simulation for solving PDEs [19–25]. In particular, some of these works provided an explicit quantum circuit with gate counts [22–25]. These results suggest that quantum algorithms have potential for accelerating the simulation of PDEs on future fault-tolerant quantum computers. However, we cannot retrieve the entire solution vector, because this needs the full state tomography; what we can do is to extract only partial information of the solution by measuring some appropriate observables [26].

Quantum algorithms are also promising for optimization problems. For discrete optimization, the variational quantum algorithms (VQAs) [27, 28] and the quantum approximate optimization algorithm (QAOA) [29–31] are among the most widely studied quantum algorithms, which are hybrid quantum–classical algorithms for combinatorial optimization. A possible drawback of these variational methods is that estimation of the gradient needs multiple running of the circuit and it can be exponentially small as the system size increases. For continuous optimization, on the other hand, quantum dynamics-based optimization [32–35] is a promising method in which an optimization problem is converted into a coherent quantum dynamics that does not involve any measurement process. In particular, quantum Hamiltonian descent (QHD) [32, 33], which maps gradient-based optimization into time-dependent Hamiltonian simulation, has the potential for offering exponential speedups under access to an objective function via bit or phase oracles [35]. However, this means that the practical use of these coherent optimization methods could be limited because such oracle construction is in general quite nontrivial and possibly hard. Moreover, even if such oracles are available, it is also uncertain whether employing QHD offers a practical advantage in terms of end-to-end gate complexity including the cost of constructing the oracle rather than query complexity.

The contribution of this paper is to develop a fully-quantum algorithm for solving PDE-constrained optimization problems, by combining the quantum PDE solver (LCHS, producing the solution $|u(t;\xi)\rangle$) and quantum optimizer (QHD, minimizing the quantized objective function $\hat{\mathcal{F}}(u(t;\xi))$ with respect to ξ). Specifically, we explicitly construct the oracle of QHD in a form of block-encoding of $\hat{\mathcal{F}}(u(t;\xi))$, in which $|u(t;\xi)\rangle$ is directly embedded without any measurement. Figure 1 illustrates the conceptual diagram of the proposed method. Importantly, the explicit block-encoding of the objective function coherently bridges a quantum PDE solver and a quantum optimizer, thereby avoiding any classical readout of the output of the quantum PDE solver $|u(t;\xi)\rangle$; the output of our algorithm is a bit string that describes the optimized solution. Hence, we can avoid the exponential overhead of classical-quantum hybrid approaches as discussed in Ref. [35]. We list our contributions as follows.

- We provide explicit constructions of block-encoding for the objective function of PDE-constrained optimization. These constructions serve as the crucial interface between two abstract quantum algorithms (i.e., the quantum PDE solver and the quantum optimizer) in practical settings, making a bottleneck-free fully-quantum algorithm.

- Based on the block-encoding constructions, we analyze the query and gate complexities of quantum algorithms for PDE-constrained optimization. This analysis clarifies the parameter dependence on the number of optimization parameters and discretization size of the underlying PDEs, thereby quantifying the potential advantages and limitations of quantum approaches. In particular, under the assumption that the optimization landscape is strongly convex, we show that there is an end-to-end quantum speedup compared to conventional classical counterparts. Such speedup is inherited from the one in quantum algorithms for solving PDEs. Specifically, our method offers a polynomial speedup for the system size and the evolution time of PDEs, and an exponential speedup for the spatial dimension of PDEs, in the regime where the system size of discretized PDEs is much larger than the number of design variables.

- We demonstrate how our framework can be applied to representative PDE-constrained optimization tasks. Specifically, we apply our framework to the parameter calibration problem in the Black-Scholes equation and the material parameter design problem in the wave equation. These results provide a pathway to exploiting quantum speedups in practical scenarios.

We remark that there have been some trials to form a constrained optimization problem from two quantum algorithms (the one for forward simulating the constraint and another for optimization). Refs [36, 37] formulated a PDE-constrained optimization, by dealing with both forward simulation and optimization in the framework of VQAs. Quantum simulation-based optimization (QuSO) algorithms [38, 39] incorporated QSVT-based forward simulation into QAOA-based optimization, where the problem Hamiltonian is constructed using forward solution, and the time evolution by the problem Hamiltonian is implemented using QSVT. As the nature of the variational method, all these methods are classical-quantum hybrid algorithms that need multiple measurements to update the optimizer and may suffer from the possible

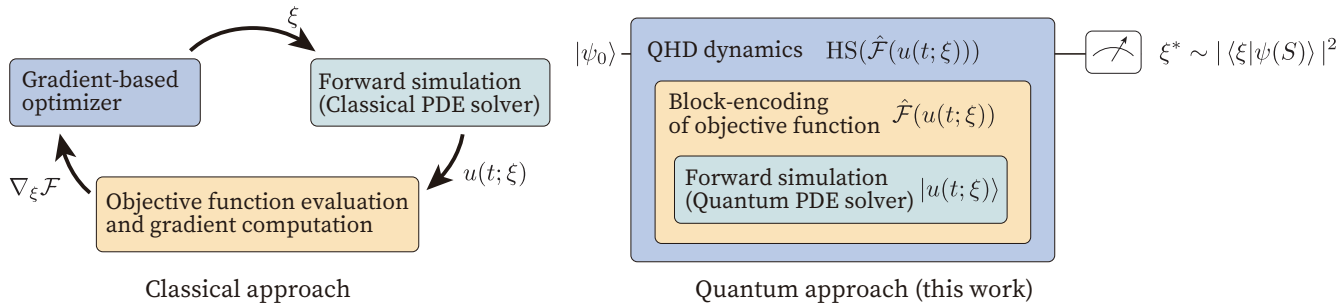

FIG. 1. Conceptual diagram of the proposed method (right), compared with the conventional classical approach (left). $\mathrm{HS}(\hat{\mathcal{F}}(u(t;\xi))) := \mathcal{T}e^{-\int_0^S (e^{-\nu s'}\nabla_\xi^2/2 + e^{\nu s'}\hat{\mathcal{F}})\mathrm{d}s'}|\psi_0\rangle$ is Hamiltonian simulation algorithm of the time-dependent Hamiltonian with the objective function as a potential term, where ν is a constant parameter.

hard-to-train issue, which does not occur in our fully-coherent solver.

The remainder of this paper is organized as follows. Section II provides some preliminaries where we formulate our problem of interest and provide primitives that compose our algorithm. Section III describes the proposed algorithm and its complexity. In Section IV, we provide two applications to demonstrate how our method can be applied to representative PDE-constrained optimization tasks. Finally, we conclude this study in Section V.

II. PRELIMINARIES

First, we provide some preliminaries on the PDE-constrained optimization, followed by the brief introduction of primitives we use: linear combination of Hamiltonian simulation [15, 16] and quantum Hamiltonian descent [32, 33]. Throughout this paper, we use the notation $|\cdot\rangle$ to describe a unit vector, which can be a quantum state embedded on a quantum computer. For example, we describe $|u(t;\xi)\rangle := u(t;\xi)/\|u(t;\xi)\|$.

A. PDE-constrained optimization

In this study, we tackle optimization problems:

$$\begin{aligned} &\min_{\xi\in[0,1]^M} && \mathcal{F}(u(t;\xi)) \\ &\text{subject to:} && \frac{\mathrm{d}u(t;\xi)}{\mathrm{d}t} = -A(\xi)u(t;\xi),\ u(0;\xi) = u_0, \end{aligned} \tag{2}$$

where we assume that the admissible space of design variables is normalized as $\Xi_{\mathrm{ad}} = [0,1]^M$, without loss of generality. The constraint is the governing equation of the system of interest, which is an ODE derived from the discretization of a PDE with $u(t) \in \mathbb{C}^N$ a discretized state variables, $u_0 \in \mathbb{C}^N$ an initial state, and $A(\xi) \in \mathbb{C}^{N\times N}$ a design-dependent coefficient matrix whose real part $\mathrm{Re}(A(\xi))$ is assumed to be positive semidefinite. Since the ODE is a linear system, we assume that $\|u_0\| = 1$ without loss of generality. The number of degrees of freedom N of the ODE is determined by the spatial discretization of PDEs. For solving a PDE on a d-dimensional domain with the characteristic length ℓ, we discretize the domain onto grids with grid spacing h_x, which makes the number of degrees of freedom N of the ODE scale as $N \in \Theta((\ell/h_x)^d)$. In this study, we do not specify the spatial discretization scheme and use the same notation u for the discretized state variables as that for the original state field $u(x,t)$ for simplicity, unless we need to distinguish them explicitly.

Here, we assume the coefficient matrix can be decomposed as follows:

$$A(\xi) := A_0 + \sum_{\mu=1}^{M} A_\mu \xi_\mu, \tag{3}$$

where $A_\mu \in \mathbb{C}^{N\times N}$ for $\mu = 1,\ldots,M$ with M the number of design variables. To ensure that $\mathrm{Re}(A(\xi))$ is positive semidefinite, we assume that $\mathrm{Re}(A_\mu)$ is positive semidefinite for all μ.

We consider the following two types of objective functions, one of which is for minimizing a quadratic form:

$$\mathcal{F}_{\mathrm{quad}}(u(\xi)) = u^\dagger(T;\xi)Pu(T;\xi), \tag{4}$$

and the other of which is for minimizing the discrepancy from the reference state $u_{\mathrm{ref}}(x)$:

$$\mathcal{F}_{\mathrm{err}}(u(\xi)) = (u(T;\xi) - u_{\mathrm{ref}})^\dagger P\,(u(T;\xi) - u_{\mathrm{ref}}), \tag{5}$$

where T is a target time, $u_{\mathrm{ref}} \in \mathbb{C}^N$ is a discretized reference state, and $P \in \mathbb{R}^{N\times N}$ is a projector. Expressing $P = \sum_{j\in\mathcal{J}} |j\rangle\langle j|$ with an index set $\mathcal{J} \subseteq [N]$, we can also express these objective functions as

$$\mathcal{F}_{\mathrm{quad}} = \sum_{j\in\mathcal{J}} \|u_j(T;\xi)\|^2, \tag{6}$$

$$\mathcal{F}_{\mathrm{err}} = \sum_{j\in\mathcal{J}} \|u_j(T;\xi) - u_{\mathrm{ref},j}\|^2, \tag{7}$$

where the subscript j represent the j-th component of the vector.

B. Quantum algorithm for ordinary differential equations

Here, we briefly introduce LCHS for simulating time-independent ODEs in Eq. (2) on a quantum computer. It should be noted that the formulation of LCHS is capable of handling time-dependent ODEs even with an inhomogeneous part. Based on the LCHS [15, 16], an analytic form of the solution for the ODE $du/dt = -Au$ with $u(0) = |u(0)\rangle$ is given as follows:

$$u(t) = \int_{\mathbb{R}} \frac{f(k)}{(1-ik)} e^{-i(H+kL)t} u(0)\ \mathrm{d}k, \tag{8}$$

where $L := (A + A^\dagger)/2$ and $H := -i(A - A^\dagger)/2$ are Hermitian operators with the requirement of L being positive semidefinite, and $f(k)$ is a kernel function satisfying certain conditions. By approximating the integration by numerical integration with discretized k, we can use LCU to prepare the state of Eq. (8) on a quantum computer as

$$\frac{1}{\alpha_{\mathrm{LCHS}}} \exp(-At)\, |u(0)\rangle |0\rangle^{\otimes a_{\mathrm{LCHS}}} + |\perp\rangle\,, \tag{9}$$

where the first term includes the solution vector $u(t) = \|u(t)\|\, |u(t)\rangle = \exp(-At)\, |u(0)\rangle$. Also, $|\perp\rangle$ represents a vector orthogonal to the first term, a_{LCHS} is the number of ancillary qubits, and α_{LCHS} is the ℓ_1 norm of the vector whose entries are kernel values $f(k)/(1-ik)$ on the discretized k space. That is, given a LCHS error ε, the LCHS can be viewed as $(\alpha_{\mathrm{LCHS}}, a_{\mathrm{LCHS}}, \varepsilon)$-block-encoding of $\exp(-At)$:

$$U_{\mathrm{LCHS}}(A) := \begin{pmatrix} \frac{\exp(-At)}{\alpha_{\mathrm{LCHS}}} & * \\ * & * \end{pmatrix}, \tag{10}$$

which we will use a subroutine for the PDE-constrained optimization.

When using the kernel function of the form $f(k) = 1/(2\pi e^{-2^\beta} e^{(1+ik)^\beta})$ for $\beta \in (0,1)$, the LCHS requires $\tilde{\mathcal{O}}((\|u(0)\|/\|u(T)\|)\alpha_A T \log(1/\varepsilon)^{1/\beta})$ queries to A (formally, queries to $H + kL$) with $\alpha_A \geq \|A\|$ to achieve error at most ε, which is nearly optimal scaling [16]. Recently, this scaling has been further improved to the optimal scaling $\tilde{\mathcal{O}}((\|u(0)\|/\|u(T)\|)\alpha_A T \log(1/\varepsilon))$ [40]. On the other hand, the simplest and original implementation of the LCHS is based on Trotterization with the kernel $f(k) = 1/(\pi(1+ik))$, which requires $\tilde{\mathcal{O}}((\|u(0)\|/\|u(T)\|)\alpha_A T^{1+1/p}/\varepsilon^{1+1/p})$ queries to Hamiltonian simulation oracles $e^{-iH\tau}$ and $e^{-iL\tau}$ where p is the order of the product formula and τ is a time increment [15]. For both cases, the factor $\|u(0)\|/\|u(T)\|$ comes from amplitude amplification. Note that in the previous paper [15], the original implementation requires additional overhead $\|u(0)\|/\|u(T)\|)^{1+1/p}$ to bound the relative Trotter error. In our study, we can omit this factor because we are interested in an unnormalized state to evaluate the objective function.

Note that the solution of the ODE is prepared as the amplitude of a quantum state. Hence, we can only extract partial information of the solution by setting some meaningful observables, which is non-trivial for practical use case.

C. Quantum Hamiltonian descent

Quantum Hamiltonian descent (QHD) is a quantum analogue of the continuous-time limit of gradient descent algorithm for continuous optimization [32, 33]. QHD encodes an objective function $\mathcal{F}(\xi)$ as the potential in the form of time-dependent (infinite-dimensional) quantum Hamiltonian $\mathcal{H}(t)$ as

$$\mathcal{H}(s) := \frac{1}{\lambda(s)} \sum_{\mu=1}^{M} \frac{1}{2} \frac{\partial^2}{\partial \xi_\mu^2} + \lambda(s)\mathcal{F}(\xi), \tag{11}$$

where s is time and $\lambda(s) > 0$ is a monotonically increasing function. Notably, the quantum state $\psi(s,\xi)$ evolved from an initial state $\psi(0,\xi)$ by the Schrödinger equation

$$\frac{\mathrm{d}\psi(s,\xi)}{\mathrm{d}t} = -i\mathcal{H}(s)\psi(s,\xi), \tag{12}$$

concentrates near low-energy regions. That is, after a sufficiently long time s, sampling ξ^* from the probability distribution $|\psi(s,\xi)|^2$ yields an approximate solution to the minimizer of $\mathcal{F}(\xi)$ with high probability. When choosing $\lambda(s) = e^{\nu s}$ with a parameter $\nu > 0$, this dynamics governed by the Schrödinger equation corresponds to the frictional dynamics with friction coefficient of ν; that is, the quantum counterpart of the classical dynamics described by Newton's equation of motion $\ddot{\xi} + \nu\dot{\xi} + \nabla\mathcal{F} = 0$.

QHD is designed to be implementable on both gate-based quantum computers and analog hardware [32]. Under the phase oracle access to the objective function, the previous work [35] showed that the quantum algorithm with $\lambda(s) = e^{\nu s}$ exhibits exponentially better scaling with the condition number of the Hessian than a vanilla gradient descent for strongly convex optimization problems. The previous work [33] also showed the exponentially fast convergence of QHD with $\lambda(s) = e^{\nu s}$ for a strongly convex function and proposed other choices of $\lambda(s)$: $\lambda(s) = s^3$ for a convex function, and $\lambda(s) = s^{1/3}$ for a non-convex function. In the work [33], the authors also provided numerical results exhibiting the better convergence of discrete-time QHD by the product formula.

An important caveat is that, to implement the QHD on a quantum computer, the oracle access to the objective function is required, and such oracle construction is in general non-trivial. This paper proposes the explicit construction of the block-encoding of the objective function in the subsequent section.

III. ALGORITHM

In this study, we propose a full quantum algorithm for solving PDE-constrained optimization problems by using QHD as an optimizer. Here, we briefly explain our proposed method, followed by the explicit block-encoding, which is an essential part of our approach and complexity analysis.

A. Overview

In QHD, we need to solve the Schrödinger equation for the time-dependent Hamiltonian $\mathcal{H}(s)$ in Eq. (11) in which the objective function of the PDE-constrained optimization is encoded into the potential term $\mathcal{F}(\xi)$. First, we discretize the Schrödinger equation so that it can be dealt with on qubits. Specifically, we discretize each design variable ξ_μ ($\mu = 1, \ldots, M$) into m bits $\xi_\mu^{(m)}, \ldots, \xi_\mu^{(1)}$, according to

$$\xi_\mu = h_\xi \sum_{b=1}^{m} \xi_\mu^{(b)} 2^{b-1}, \tag{13}$$

where h_ξ is the interval of discretized design variables, i.e., $h_\xi = 1/(2^m - 1)$. Hereinafter, we use the notation

$$|\xi\rangle = |\xi_M, \ldots, \xi_1\rangle = \left|\xi_M^{(m)}, \ldots, \xi_M^{(1)}, \ldots, \xi_1^{(m)}, \ldots, \xi_1^{(1)}\right\rangle \tag{14}$$

to represent the design variables ξ encoded in the computational basis of a Mm-qubit system. The quantum state $\psi(s, \xi)$ is represented in terms of the ket vector on the computational basis as $|\psi(s)\rangle = \sum_\xi \psi(s, \xi) |\xi\rangle$. By this discretization, the second-order derivative $\partial^2/\partial\xi_\mu^2$ in the Hamiltonian is replaced by the finite difference operator $\Lambda \in \mathbb{R}^{2^m \times 2^m}$ defined as

$$\Lambda := \frac{1}{h_\xi^2} \begin{pmatrix} -2 & 1 & & & \\ 1 & -2 & 1 & & \\ & \ddots & \ddots & \ddots & \\ & & 1 & -2 & 1 \\ & & & 1 & -2 \end{pmatrix}. \tag{15}$$

Thus, the discrete Laplace operator for the Laplacian $\sum_{\mu=1}^{M} \partial^2/\partial\xi_\mu^2$ is replaced with

$$\Lambda^{(M)} := \sum_{\mu=1}^{M} I^{\otimes(M-\mu)} \otimes \Lambda \otimes I^{\otimes(\mu-1)}, \tag{16}$$

where I is an identity matrix of size 2×2. The time-dependent Hamiltonian after discretization is then obtained, as follows:

$$\mathcal{H}(s) = \frac{1}{2\lambda(s)} \Lambda^{(M)} + \lambda(s) \hat{\mathcal{F}}, \tag{17}$$

where $\hat{\mathcal{F}} := \sum_\xi \mathcal{F}(u(\xi)) |\xi\rangle\langle\xi|$. In the present study, we choose $\lambda(s) = e^{\nu s}$.

The proposed method for PDE-constrained optimization consists of the following steps.

1. Set the friction parameter ν and the simulation time S.

2. Prepare an initial state $|\psi(0)\rangle = U_{\psi_0} |0\rangle^{\otimes Mm}$ representing the initial probability distribution of design variables ξ by a state preparation oracle U_{ψ_0}. This study uses the uniform superposition as the initial state, i.e., $|\psi(0)\rangle = 2^{-Mm/2} \sum_\xi |\xi\rangle$, which can be prepared by the Hadamard gate.

3. Encode the PDE solution $|u(t)\rangle$ coherently into the objective function of the PDE-constrained optimization, $\mathcal{F}(\xi)$; then, perform the time-dependent Hamiltonian simulation of $\mathcal{H}(s)$ in Eq. (17) from time 0 to S.

4. Obtain a solution $\xi^* \sim |\langle\xi|\psi(S)\rangle|^2$ by measurement on the computational basis.

The key component of the QHD is the step 3, i.e. the implementation of the time evolution of the time-dependent Hamiltonian $\mathcal{H}(s)$, which contains the PDE solution $|u(t)\rangle$. To implement the time evolution $U_{\mathcal{H}}(s, 0) := \mathcal{T} e^{-i \int_0^s \mathcal{H}(s')\mathrm{d}s'}$ where $\mathcal{T}$ is a time-ordering operator, we employ two approaches:

- the interaction picture simulation using a truncated Dyson series;

- a simple product formula.

The former is effective for accurately simulating the time-dependent Hamiltonian, which ensures convergence to the optimal solution for strongly convex-optimization problems, while the latter provides a simple implementation despite an unfavorable error. In either case, we divide the time interval into short ones $0 := s_0 < s_1 < \cdots < s_{N_{\mathrm{step}}-1} < s_{N_{\mathrm{step}}} =: S$ with uniform interval Δs, i.e., $s_j = j\Delta s$.

In the interaction picture simulation, we transform the time-dependent Hamiltonian $\mathcal{H}(s)$ into the interaction picture one:

$$\begin{aligned} \mathcal{H}_{\mathrm{I}}(s) &:= e^{i \int_0^s e^{\nu s'} \hat{\mathcal{F}} \mathrm{d}s'} \frac{e^{-\nu s}}{2} \Lambda^{(M)} e^{-i \int_0^s e^{\nu s'} \hat{\mathcal{F}} \mathrm{d}s'} \\ &= e^{i \frac{1}{\nu}(e^{\nu s} - 1) \hat{\mathcal{F}}} \frac{e^{-\nu s}}{2} \Lambda^{(M)} e^{-i \frac{1}{\nu}(e^{\nu s} - 1) \hat{\mathcal{F}}}, \end{aligned} \tag{18}$$

where the quantum state is evolved as

$$|\psi(s)\rangle = \mathcal{T} e^{-i \int_0^s e^{\nu s'} \hat{\mathcal{F}} \mathrm{d}s'} \; \mathcal{T} e^{-i \int_0^s \mathcal{H}_I(s')\mathrm{d}s'} |\psi(0)\rangle. \tag{19}$$

Then we approximate the time evolution $\mathcal{T} e^{-i \int_0^s \mathcal{H}_I(s')\mathrm{d}s'}$ by the multiple short-time evolution from time s_j to s_{j+1}, denoted by $U_{\mathrm{TDS}}(s_{j+1}, s_j) := \mathcal{T} e^{-i \int_{s_j}^{s_{j+1}} \mathcal{H}_{\mathrm{I}}(s')\mathrm{d}s'}$, by the

truncated Dyson series [41]. The total approximated time evolution $U_{\mathcal{H}}(S,0)$ is given as

$$U_{\mathcal{H}}(S,0) \approx e^{-i\frac{1}{\nu}(e^{\nu S}-1)\hat{\mathcal{F}}} U_{\mathrm{TDS}}(t_{N_{\mathrm{step}}}, t_{N_{\mathrm{step}}-1}) \cdots U_{\mathrm{TDS}}(t_1, t_0). \tag{20}$$

In the simple product formula, we approximate the short-time evolution of the Hamiltonian $\mathcal{H}(s)$ [33] as

$$U_{\mathcal{H}}(s_{j+1}, s_j) \approx e^{-i\frac{1}{2}e^{-\nu s_j}\Lambda^{(M)}\Delta s} e^{-ie^{\nu s_j}\hat{\mathcal{F}}\Delta s} =: U_{\mathrm{PF}}(s_{j+1}, s_j). \tag{21}$$

Then, we approximate the time evolution $U_{\mathcal{H}}(S,0)$, as

$$U_{\mathcal{H}}(S,0) \approx U_{\mathrm{PF}}(t_{N_{\mathrm{step}}}, t_{N_{\mathrm{step}}-1}) \cdots U_{\mathrm{PF}}(t_1, t_0). \tag{22}$$

In either case, we need to implement the time evolution of the objective function $e^{-i\hat{\mathcal{F}}s}$ for an arbitrary time increment s. In the interaction picture simulation, we need to implement the Hamiltonian by some means; here we employ the block-encoding of $\mathcal{H}_I(s)$, as an access model. For the product formula, we need the time evolution of the discrete Laplace operator $\Lambda^{(M)}$, which is given in Appendix A.

Our complexity analysis focuses on the interaction picture simulation under the strong-convexity assumption, even though objective functions of PDE-constrained optimization are often not strongly convex. When strong convexity is absent, such convergence guarantees can disappear, and the practical value of executing a highly accurate time-dependent Hamiltonian simulation may diminish. This motivates a more heuristic stance: rather than expending resources to suppress simulation error, we are interested in whether a coarse but simple time evolution is sufficient for the optimizer to provide useful optimized solutions. This is the reason why we consider the simple product formula as a minimal implementation, which is numerically examined later. In short, we use the truncated Dyson series to derive explicit complexity bounds for the strongly convex case, and we use a product formula as a simple baseline whose practical value we confirm by experiment.

B. Block-encoding of the objective function for time evolution

As mentioned in the previous subsection, our primal problem is how to implement the time evolution of the objective function $e^{-i\hat{\mathcal{F}}s}$, where again $\hat{\mathcal{F}} = \sum_\xi \mathcal{F}(u(\xi))\,|\xi\rangle\langle\xi|$. Here, we construct a block-encoding of the objective function $\mathcal{F}(u(\xi))$ for implementing its time evolution. Our goal is to implement a unitary U_{obj} which satisfies

$$(I^{\otimes Mm} \otimes \langle 0|^{\otimes a_{\mathrm{obj}}}) U_{\mathrm{obj}} (I^{\otimes Mm} \otimes |0\rangle^{\otimes a_{\mathrm{obj}}}) = \frac{1}{\alpha_{\mathrm{obj}}}\left(\hat{\mathcal{F}} - \delta_{\mathrm{obj}} I^{\otimes Mm}\right), \tag{23}$$

where a_{obj} is the number of ancillary qubits for the block-encoding, α_{obj} is a normalization constant so that the objective function can be encoded into a unitary, and δ_{obj} is an offset. As a first step, we first aim at implementing a design-dependent forward simulation oracle U_{for} that satisfies

$$U_{\mathrm{for}}\,|\xi\rangle|0\rangle^{\otimes n}|0\rangle^{\otimes a_{\mathrm{for}}} = \frac{\|u(t;\xi)\|}{\alpha_{\mathrm{for}}}\,|\xi\rangle|u(t;\xi)\rangle|0\rangle^{\otimes a_{\mathrm{for}}} + \sqrt{1-\frac{\|u(t;\xi)\|^2}{\alpha_{\mathrm{for}}^2}}\,|\xi\rangle|\chi\rangle\,, \tag{24}$$

as a direct extension of Eq. (9) so that the space of design parameters are included in the dynamics. Here, $|\chi\rangle$ is a state orthogonal to the state $|u(t;\xi)\rangle|0\rangle^{\otimes a_{\mathrm{for}}}$, a_{for} is the number of ancillary qubits, $n := \log_2 N$, and α_{for} is a constant for normalization. Also, $|u(t;\xi)\rangle := (1/\|u(t;\xi)\|)\sum_{j=0}^{2^n-1} u_j(t;\xi)\,|j\rangle$ is the normalized solution to the linear equation (2), represented in terms of the computational basis $\{|j\rangle\}$. To this end, we consider the following ODE for an extended state variable $\tilde{u}(t) \in \mathbb{C}^{2^{(Mm+n)}}$:

$$\frac{\mathrm{d}\tilde{u}(t)}{\mathrm{d}t} = -\tilde{A}(\xi)\tilde{u}(t), \tag{25}$$

where $\tilde{A}(\xi) := \sum_\xi (|\xi\rangle\langle\xi| \otimes A(\xi))$. This dynamics evolves an initial state $|\tilde{u}(0)\rangle := |\psi\rangle|u_0\rangle$ with arbitrary quantum state $|\psi\rangle = \sum_\xi \psi_\xi\,|\xi\rangle \in \mathbb{C}^{2^{Mm}}$. Since the analytic form of the solution for this ODE is given as

$$\begin{aligned}\tilde{u}(t) &= e^{-\tilde{A}(\xi)t}\tilde{u}(0)\\ &= \prod_\xi \left(|\xi\rangle\langle\xi| \otimes e^{-A(\xi)t} + \left(I^{\otimes Mm} - |\xi\rangle\langle\xi|\right) \otimes I^{\otimes n}\right)\tilde{u}(0)\\ &= \sum_\xi \psi_\xi\,|\xi\rangle \otimes \|u(t;\xi)\|\,|u(t;\xi)\rangle\,,\end{aligned} \tag{26}$$

the forward simulation oracle can be implemented using the LCHS oracle $U_{\mathrm{LCHS}}(\tilde{A})$ with a state preparation oracle U_{u_0} acting as $|u_0\rangle = U_{u_0}|0\rangle^{\otimes n}$; that is, $U_{\mathrm{for}} = U_{\mathrm{LCHS}}(\tilde{A})(I^{\otimes Mm} \otimes U_{u_0} \otimes I^{\otimes a_{\mathrm{for}}})$. To implement the LCHS oracle $U_{\mathrm{LCHS}}(\tilde{A})$ for the design-dependent matrix $\tilde{A}$, we use the following relation:

$$\begin{aligned}\tilde{A} &= \sum_\xi (|\xi\rangle\langle\xi| \otimes A(\xi))\\ &= I^{\otimes Mm} \otimes A_0\\ &\quad + \sum_{\mu=1}^{M}\sum_{b=1}^{m}\sum_{\xi_\mu^{(m)},\ldots,\xi_\mu^{(1)}=0}^{1} h_\xi 2^{b-1}\left(|\xi\rangle\langle\xi| \otimes A_\mu \xi_\mu^{(b)}\right)\\ &= I^{\otimes Mm} \otimes A_0 + \sum_{\mu=1}^{M}\sum_{b=1}^{m} h_\xi 2^{b-1}\left(|1\rangle\langle 1|_{\mu,b} \otimes A_\mu\right),\end{aligned} \tag{27}$$

where $|1\rangle\langle 1|_{\mu,b}$ acts on the b-th qubit for the μ-th design variable. Thus, a block-encoding of $\tilde{A}$ can be obtained via linear combination of block-encoded matrices of A_μ [42, Lemma 52], using $\lceil \log_2(Mm) \rceil$ additional qubits with the ℓ_1-norm of the coefficients equal to $M+1$. Based on Eqs. (10) and (24), we obtain the relation of $\alpha_{\rm for} = \alpha_{\rm LCHS}$ and $a_{\rm for} = a_{\rm LCHS} + \lceil \log_2(Mm) \rceil$.

We are now ready to construct block-encodings of the objective functions given by Eqs. (4) and (5). Recall that both these quantities are defined in terms of the projector $P = \sum_{j\in\mathcal{J}} |j\rangle\langle j|$. Then, by applying the reflection $R_P := 2P \otimes |0\rangle\langle 0|^{\otimes a_{\rm for}} - I$, where $|0\rangle\langle 0|^{\otimes a_{\rm for}}$ is for the success flag of LCHS, we have

$$\begin{aligned}
&(I^{\otimes Mm} \otimes R_P) U_{\rm for} |\xi\rangle |0\rangle^{\otimes n} |0\rangle^{\otimes a_{\rm for}} \\
&= \frac{\|u(t;\xi)\|}{\alpha_{\rm for}} |\xi\rangle \left(2P - I^{\otimes n}\right) |u(t;\xi)\rangle |0\rangle^{\otimes a_{\rm for}} \\
&\quad - \sqrt{1 - \frac{\|u(t;\xi)\|^2}{\alpha_{\rm for}^2}} |\xi\rangle |\chi\rangle . \qquad (28)
\end{aligned}$$

Using this equation, we can construct a unitary $U_{\rm obj,quad} := U_{\rm for}^\dagger (I^{\otimes Mm} \otimes R_P) U_{\rm for}$ which is the $(\alpha_{\rm for}^2/2, n + a_{\rm for}, 2\alpha_{\rm for}\varepsilon)$-block-encoding of the objective function $\mathcal{F}_{\rm quad}$ as follows:

$$\begin{aligned}
&\left(I^{\otimes Mm} \otimes \langle 0|^{\otimes (n+a_{\rm for})}\right) U_{\rm for}^\dagger R_P U_{\rm for} \left(I^{\otimes Mm} \otimes |0\rangle^{\otimes (n+a_{\rm for})}\right) \\
&= \sum_{\xi,\xi'} \left(|\xi'\rangle\langle\xi'| \otimes \langle 0|^{\otimes (n+a_{\rm for})}\right) U_{\rm for}^\dagger R_P U_{\rm for} \left(|\xi\rangle\langle\xi| \otimes |0\rangle^{\otimes (n+a_{\rm for})}\right) \\
&= \sum_{\xi} \left(\frac{\|u(t;\xi)\|^2}{\alpha_{\rm for}^2} \langle u(t;\xi)| \left(2P - I^{\otimes n}\right) |u(t;\xi)\rangle - 1 + \frac{\|u(t;\xi)\|^2}{\alpha_{\rm for}^2} \right) |\xi\rangle\langle\xi| \\
&= \sum_{\xi} \left(\frac{2\|u(t;\xi)\|^2}{\alpha_{\rm for}^2} \langle u(t;\xi)| P |u(t;\xi)\rangle - 1 \right) |\xi\rangle\langle\xi| \\
&= \frac{2\hat{\mathcal{F}}_{\rm quad}(u(\xi))}{\alpha_{\rm for}^2} - 1, \qquad (29)
\end{aligned}$$

where we regard $|0\rangle^{\otimes n}$ as ancilla qubits and the error ε comes from the error of LCHS. We use the formula [42, Lemma 53] to derive the error of the block-encoding $2\alpha_{\rm for}\varepsilon$. Figure 2 illustrates the corresponding quantum circuit. This block-encoding corresponds to our goal in Eq. (23) with $\alpha_{\rm obj} = \delta_{\rm obj} = \alpha_{\rm for}^2/2$.

Next, for encoding $\mathcal{F}_{\rm err}$, we assume a reference state preparation oracle $U_{\rm ref}$ that prepares

$$U_{\rm ref} |0\rangle^{\otimes n} = |u_{\rm ref}\rangle , \qquad (30)$$

where $|u_{\rm ref}\rangle = u_{\rm ref}/\|u_{\rm ref}\|$ is a normalized reference state that can be encoded into a quantum state. Then, we consider LCU of $\alpha_{\rm for} U_{\rm for} - \|u_{\rm ref}\| U_{\rm ref}$ by a preparation oracle $U_{\rm PREP}$ acting as

$$U_{\rm PREP} |0\rangle = \frac{1}{\sqrt{\alpha_{\rm for} + \|u_{\rm ref}\|}} \left(\sqrt{\alpha_{\rm for}} |0\rangle + \sqrt{\|u_{\rm ref}\|} |1\rangle \right), \qquad (31)$$

and a select oracle $U_{\rm SEL}$ defined as

$$U_{\rm SEL} := U_{\rm for} \otimes |0\rangle\langle 0| + I^{\otimes Mm} \otimes U_{\rm ref} \otimes I^{\otimes a_{\rm for}} \otimes |1\rangle\langle 1| , \qquad (32)$$

which prepares the quantity $u(t;\xi) - u_{\rm ref}$ as follows.

$$\begin{aligned}
&(I^{\otimes (Mm+n+a_{\rm for})} \otimes U_{\rm PREP}^\dagger)(I^{\otimes (Mm+n+a_{\rm for})} \otimes Z) U_{\rm SEL} (I^{\otimes (Mm+n+a_{\rm for})} \otimes U_{\rm PREP}) |\xi\rangle |0\rangle^{\otimes (n+a_{\rm for}+1)} \\
&= \left(\frac{1}{\alpha_{\rm LCU}} \left(\alpha_{\rm for} U_{\rm for} - \|u_{\rm ref}\| I^{\otimes Mm} \otimes U_{\rm ref} \otimes I^{\otimes a_{\rm for}} \right) |\xi\rangle |0\rangle^{\otimes (n+a_{\rm for})} \right) |0\rangle + |\perp\rangle \\
&= \frac{1}{\alpha_{\rm LCU}} |\xi\rangle (\|u(t;\xi)\| \, |u(t;\xi)\rangle - \|u_{\rm ref}\| \, |u_{\rm ref}\rangle) |0\rangle^{\otimes a_{\rm for}+1} \\
&\quad + |\perp\rangle , \qquad (33)
\end{aligned}$$

where $\alpha_{\rm LCU} = \alpha_{\rm for} + \|u_{\rm ref}\|$. With this unitary denoted by $U_{\rm LCU} := (I \otimes U_{\rm PREP}^\dagger)(I \otimes Z) U_{\rm SEL} (I \otimes$

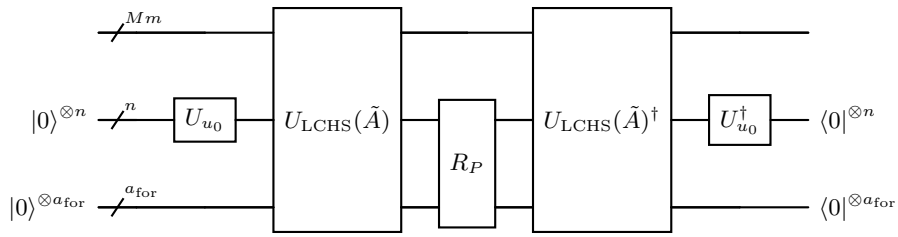

FIG. 2. Quantum circuit for block-encoding of $\mathcal{F}_{\mathrm{quad}}$.

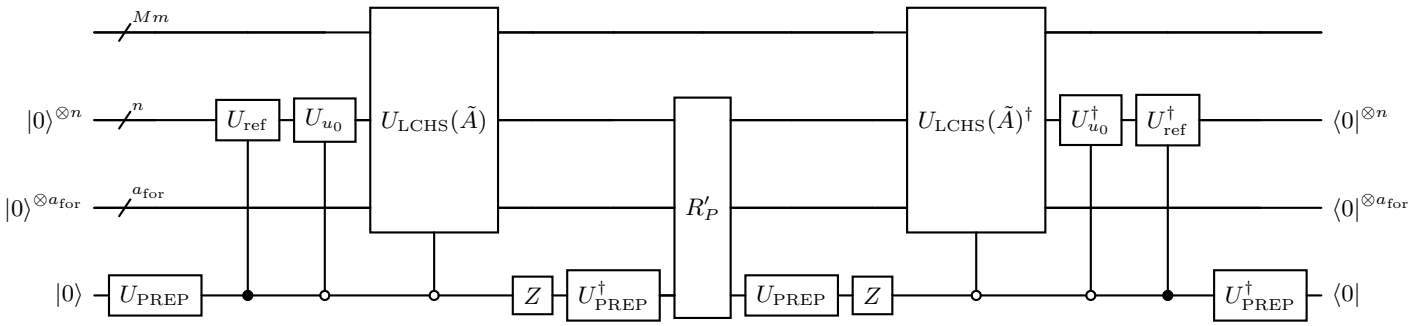

FIG. 3. Quantum circuit for block-encoding of $\mathcal{F}_{\mathrm{err}}$.

U_{PREP}), and a reflection operator $R'_P := 2P \otimes |0\rangle\langle 0|^{\otimes(a_{\mathrm{for}}+1)} - I^{\otimes(n+a_{\mathrm{for}}+1)}$, we construct a unitary $U_{\mathrm{obj,err}} := U_{\mathrm{LCU}}^\dagger (I^{\otimes Mm} \otimes R'_P) U_{\mathrm{LCU}}$ which is the $(\alpha_{\mathrm{LCU}}^2/2, n+a_{\mathrm{for}}+1, 2\alpha_{\mathrm{LCU}}\varepsilon)$-block-encoding the objective function $\mathcal{F}_{\mathrm{err}}$, as follows:

$$
\begin{aligned}
&\left(I^{\otimes Mm} \otimes \langle 0|^{\otimes(n+a_{\mathrm{for}}+1)}\right) U_{\mathrm{LCU}}^\dagger (I^{\otimes Mm} \otimes R'_P) U_{\mathrm{LCU}} \left(I^{\otimes Mm} \otimes |0\rangle^{\otimes(n+a_{\mathrm{for}}+1)}\right) \\
&= \sum_{\xi,\xi'} \left(|\xi'\rangle\langle\xi'| \otimes \langle 0|^{\otimes(n+a_{\mathrm{for}}+1)}\right) U_{\mathrm{LCU}}^\dagger (I^{\otimes Mm} \otimes R'_P) U_{\mathrm{LCU}} \left(|\xi\rangle\langle\xi| \otimes |0\rangle^{\otimes(n+a_{\mathrm{for}}+1)}\right) \\
&= \sum_{\xi,\xi'} \left(\frac{2\left(u(t;\xi) - u_{\mathrm{ref}}\right)^\dagger P \left(u(t;\xi) - u_{\mathrm{ref}}\right)}{\alpha_{\mathrm{LCU}}^2} - 1 \right) |\xi\rangle\langle\xi| \\
&= \frac{2\hat{\mathcal{F}}_{\mathrm{err}}(\xi)}{\alpha_{\mathrm{LCU}}^2} - 1,
\end{aligned}
\tag{34}
$$

where we use the formula [42, Lemma 53] to derive the error of the block-encoding $2\alpha_{\mathrm{LCU}}\varepsilon$. Figure 3 illustrates the corresponding quantum circuit. This block-encoding corresponds to that in Eq. (23) with $\alpha_{\mathrm{obj}} = \delta_{\mathrm{obj}} = \alpha_{\mathrm{LCU}}^2/2$. Now, by QSVT combined with oblivious amplitude amplification (OAA) [11], we obtain the time evolution $e^{-i\hat{\mathcal{F}}t}$ for both cases $\hat{\mathcal{F}} = \hat{\mathcal{F}}_{\mathrm{quad}}$ and $\hat{\mathcal{F}} = \hat{\mathcal{F}}_{\mathrm{err}}$.

C. Complexity analysis

In this subsection, we derive the query and gate complexities of the block-encoding of the objective function, followed by the total gate complexity of our algorithm. Then, we compare it with that of a classical counterpart. We also discuss the space complexity of our algorithm.

1. *Gate complexity of block-encoding*

Here, we derive the gate complexity for the block-encoding of the objective functions $\hat{\mathcal{F}}$. First, we derive the query complexity of the objective function to the coefficient matrices A_μ. Since the block-encoding of the objective function requires a single use of $U_{\mathrm{LCHS}}(\tilde{A})$ and $U_{\mathrm{LCHS}}(\tilde{A})^\dagger$ for $\mathcal{F}_{\mathrm{quad}}$, or their controlled version for $\mathcal{F}_{\mathrm{err}}$, it suffices to evaluate the query complexity to A_μ for im-

plementing $U_{\mathrm{LCHS}}(\tilde{A})$.

Based on the previous work [16], $U_{\mathrm{LCHS}}(\tilde{A})$ without amplitude amplification requires $\tilde{\mathcal{O}}(\tilde{\alpha}_A T \log(1/\varepsilon)^{1/\beta})$ queries to $\tilde{A}$ where $\tilde{\alpha}_A \geq \|\tilde{A}\|$. Note that our method does not include amplitude amplification because the objective function requires an unnormalized vector $u(t;\xi)$. Since Eq. (27) means that $\|\tilde{A}\| \leq (M+1)(\|A_0\| + \max_\mu \|A_\mu\|)$, the query complexity of $U_{\mathrm{LCHS}}(\tilde{A})$ results in $\tilde{\mathcal{O}}(\alpha_A M T \log(1/\varepsilon)^{1/\beta})$ with $\alpha_A \geq \|A_0\| + \max_\mu \|A_\mu\|$, which is the query complexity for the block-encoding of the objective functions.

When we assume that PDEs are defined on the d-dimensional space, the block-encoding of a finite difference operator defined in Appendix A requires $\mathcal{O}(n^2/d)$ single and two-qubit gates, and its norm is $\mathcal{O}(d/h_x^\eta)$ with h the interval of spatial discretization, and η the order of spatial derivative in the coefficient matrices, as discussed in Appendix A. Assuming that coefficients of PDEs can be block-encoded by $\mathcal{O}(n)$ gates and the dominant coefficient of the PDE is α_{PDE} (see Appendix B), we can obtain the block-encoding of coefficient matrices with $\alpha_A \in \mathcal{O}(\alpha_{\mathrm{PDE}} d/h_x^\eta)$ using $\mathcal{O}(n^2/d + n)$ gates. Furthermore, the block-encoding of $\tilde{A}$ in Eq. (27) requires Mm times applications of block-encodings of A_μ.

Other than queries to the coefficient matrices, the block-encoding of the objective function includes

1. Two uses of state preparation of initial forward solution by unitary U_{u_0} for $|u_0\rangle = U_{u_0}|0\rangle$. We assume that the initial state of forward simulation u_0 can be efficiently prepared so that U_{u_0} can be implemented using $\mathcal{O}(\mathrm{polylog}(N))$ single and two-qubit gates.

2. A single use of the reflection operator R_P or R'_P. Here, we assume that the projector P is represented as $P = P_n \otimes \cdots \otimes P_1$ for $P_\cdot \in \{|0\rangle\langle 0|, |1\rangle\langle 1|, I\}$. Then we can implement R_P using at most $(n + a_{\mathrm{for}})$-controlled Z gate, which is decomposed into $\mathcal{O}(n + a_{\mathrm{for}})$ single and two-qubit gates.

3. Two uses of reference state preparation by unitary U_{ref} for $|u_{\mathrm{ref}}\rangle = U_{\mathrm{ref}}|0\rangle$ when the objective function has the form of $\mathcal{F}_{\mathrm{err}}$. We assume that the reference state can be efficiently prepared so that U_{ref} can be implemented using $\mathcal{O}(\mathrm{polylog}(N))$ single and two-qubit gates.

Thus, the gate complexity of the block-encoding of the objective function results in

$$\tilde{\mathcal{O}}\left(\frac{\alpha_{\mathrm{PDE}}(n^2+dn)M^2 mT}{h_x^\eta}\log(1/\varepsilon)^{1/\beta} + \mathrm{polylog}(N)\right). \tag{35}$$

When PDEs are nondimensionalized, which means that $h_x \sim \mathcal{O}(1/N^{1/d})$, we obtain the gate complexity of

$$\tilde{\mathcal{O}}\left(\alpha_{\mathrm{PDE}} d N^{\frac{\eta}{d}} M^2 m T \log(1/\varepsilon)^{1/\beta}\right), \tag{36}$$

where $\tilde{\mathcal{O}}$ hinders poly-logarithmic factors in all parameters other than ε.

In the case of the simple implementation by Trotterization, a similar calculation derives the query complexity for the block-encoding of the objective function as $\tilde{\mathcal{O}}(\alpha_A M T^{1+1/p}/\varepsilon^{1+1/p})$ where p is the order of the product formula. This means that the gate complexity is $\tilde{\mathcal{O}}(\alpha_{\mathrm{PDE}} d M^2 m N^{\eta/d} T^{1+1/p}/\varepsilon^{1+1/p})$.

2. *Gate complexities of our algorithm*

In addition to the query to any of the coefficient matrices, our algorithm includes several other components. Here, we describe these components and derive the gate complexity for our algorithm to simulate the time evolution of $\mathcal{H}(s)$ within an error ϵ. First, we define the key component $\texttt{HAM-T}_j$, which is a block-encoding of

$$\sum_{q=0}^{Q-1} |q\rangle\langle q| \otimes \frac{1}{\alpha_\Lambda}\mathcal{H}_{\mathrm{I}}\left(s_j + \frac{q\Delta s}{Q}\right), \tag{37}$$

where $Q \in \mathcal{O}(\frac{S}{\epsilon}(\alpha_\Lambda + \alpha_{\mathrm{obj}} e^{\nu S} + \nu))$, and $\alpha_\Lambda = 5M/2h_\xi^2$, requiring $a_{\mathrm{HT}} = 4 + 2a_{\mathrm{obj}} + 2\lceil \log\big((1/\nu)(16\alpha_\Lambda N_{\mathrm{step}} e^{\nu S})/\epsilon\big)\rceil + \lceil\log(5M)\rceil + \lceil\log(Q)\rceil$ ancillary qubits, as explained in Ref. [35, 41] and Appendix C. The components we use for our algorithm are as follows.

1. A single use of state preparation of design variables by unitary U_{ψ_0} for $|\psi(0)\rangle = U_{\psi_0}|0\rangle^{\otimes Mm}$. We use Mm Hadamard gates to prepare the state $|\psi(0)\rangle = 2^{-Mm/2}\sum_\xi |\xi\rangle$.

2. $N_{\mathrm{step}} \in \mathcal{O}(\alpha_\Lambda S)$ segments of interaction picture simulation by truncated Dyson series. As described in Ref. [35] , this part requires

   (a) A single execution of $e^{-i\frac{1}{\nu}(e^{\nu S}-1)\hat{\mathcal{F}}}$,

   (b) $\mathcal{O}(\alpha_\Lambda S \frac{\log(\alpha_\Lambda S/\epsilon)}{\log\log(\alpha_\Lambda S/\epsilon)})$ queries to $\texttt{HAM-T}_j$ oracle,

   (c) $\mathcal{O}\left(\alpha_\Lambda S a_{\mathrm{HT}} \frac{\log(\alpha_\Lambda S/\epsilon)}{\log\log(\alpha_\Lambda S/\epsilon)}\right)$ primitive gates.

Since the number of quantum gates for the first component is negligible compared with that for the second one, we focus on the gate complexity of the second component. The number of single and two-qubit gates required for each component (a), (b), and (c) is, respectively,

$$\text{(a):}\qquad \tilde{\mathcal{O}}\left(\frac{1}{\nu}e^{\nu S}\alpha_{\rm PDE}dN^{\frac{\eta}{d}}M^2mT\log\left(\frac{e^{\nu S}/\nu}{\epsilon}\right)^{1+\frac{1}{\beta}}\right), \tag{38}$$

$$\text{(b):}\qquad \tilde{\mathcal{O}}\left(\left(\frac{1}{\nu}e^{\nu S}\alpha_{\rm PDE}dN^{\frac{\eta}{d}}M^2mT\log\left(\frac{e^{\nu S}/\nu}{h_\xi^4\epsilon}\right)^{2+\frac{1}{\beta}}+\log\left(\frac{M/h_\xi^2+e^{\nu S}+\nu}{\epsilon}\right)\right)\frac{MS}{h_\xi^2}\log\left(\frac{1}{\epsilon}\right)\right), \tag{39}$$

$$\text{(c):}\qquad \tilde{\mathcal{O}}\left(\frac{MS}{h_\xi^2}\left(\log\left(\frac{e^{\nu S}+\nu}{\epsilon}\right)^2+\log\left(\frac{e^{\nu S}/\nu}{\epsilon}\right)^2\right)\right). \tag{40}$$

where we regard $\alpha_{\rm obj}\in\mathcal{O}(1)$, which is problem-independent, and $\tilde{\mathcal{O}}$ hinders poly-logarithmic factors in all parameters. Obviously, the component (b) is dominant and is the total gate complexity of our algorithm. Although it is difficult to predetermine an appropriate optimization time S in general for PDE-constrained optimization, we restrict the time to $S\sim\mathcal{O}(\log(1/\epsilon)/\nu)$, which ensures that the optimized solution ξ^* satisfies $|\mathcal{F}(u(\xi^*))-\min_\xi\mathcal{F}(u(\xi))|\le\epsilon$ when the objective function is strongly convex [35]. Then, we obtain the total gate complexity of our algorithm as

$$\tilde{\mathcal{O}}\left(\frac{\alpha_{\rm PDE}dN^{\frac{\eta}{d}}M^3T}{h_\xi^2\epsilon}\right), \tag{41}$$

where we regard ν as a constant parameter.

One limitation of this analysis is that we can ensure the convergence when the objective function is strongly convex. In general cases, the objective function can often be non-convex, where we may not need to focus on the precise simulation of the time evolution by the time-dependent Hamiltonian $\mathcal{H}(s)$.

3. *Classical counterpart*

Conventional classical approaches with a p-th-order time integrator requires $\mathcal{O}(\alpha_A^{1+\frac{1}{p}}T^{1+\frac{1}{p}}\epsilon^{-1/p})$ times of matrix-vector multiplication $A(\xi)u$ for forward simulation within an error ϵ for the objective function evaluation. Under the reasonable assumption that all A_μ are sparse matrices in discretizing PDEs, these classical approaches for forward simulation require

$$\mathcal{O}\left(MN+\frac{N\alpha_{\rm PDE}^{1+\frac{1}{p}}d^{1+\frac{1}{p}}N^{\frac{\eta}{d}(1+\frac{1}{p})}T^{1+\frac{1}{p}}}{\epsilon^{1/p}}\right) \tag{42}$$

times arithmetic operations where the first term is for constructing $A(\xi)$ and the second term for time integration. To compute gradients of the objective function, it is reasonable to employ the adjoint variable method. In the method, one solves the adjoint problem, which requires the same complexity as the forward simulation, and then computes the gradients by $\mathcal{O}(MNT)$ additional arithmetic operations, which are used to evaluate the inner product of the forward and adjoint solutions. That is, the computational complexity for one iteration of optimization is $\mathcal{O}(MNT+\alpha_{\rm PDE}^{1+1/p}d^{1+1/p}N^{(\eta/d)(1+1/p+d/\eta)}T^{1+1/p}/\epsilon^{1/p})$. Taking the same order of optimization steps $\mathcal{O}(\log(1/\epsilon))$ as that for our algorithm, we obtain

$$\tilde{\mathcal{O}}\left(MNT\log\left(\frac{1}{\epsilon}\right)+\frac{\alpha_{\rm PDE}^{1+1/p}d^{1+\frac{1}{p}}N^{\frac{\eta}{d}(1+\frac{1}{p}+\frac{d}{\eta})}T^{1+\frac{1}{p}}}{\epsilon^{1/p}}\right). \tag{43}$$

Therefore, by the comparison between Eq. (41) and Eq. (43) in the regime $N\gg M$, our algorithm can provide a polynomial speedup of degree $1+d/\eta+1/p$ for the system size of PDEs, N, an exponential speedup for the spatial dimension of PDEs, d, and a polynomial speedup of degree $1+1/p$ for the evolution time of PDEs, T. These speedups are inherited from those in the forward PDE simulation. This result implies that our algorithm is suitable for problems requiring large-scale and high-dimensional simulations with a moderate number of parameters.

4. *Space complexity*

The number of qubits required for our algorithm is $Mm+a_{\rm HT}$ where Mm and $a_{\rm HT}$ are the number of system and ancillary qubits for `HAM-T`$_j$, respectively. Given that the $a_{\rm LCHS}$ ancillary qubits for LCHS are $a_{\rm LCHS}=\mathcal{O}(\log(\tilde{\alpha}_A T\log(1/\varepsilon)^{1+1/\beta}))=\mathcal{O}(\log(\alpha_{\rm PDE}dN^{\eta/d}MT\log(1/\varepsilon)^{1+1/\beta}))$ [16], the total number of qubits scales as

$$M\lceil\log_2(1/h_\xi+1)\rceil+\tilde{\mathcal{O}}\left(\frac{\eta}{d}\log N+\log\left(\frac{\alpha_{\rm PDE}dMT}{h_\xi\epsilon}\right)\right), \tag{44}$$

where we explicitly describe Mm qubits for design variables.

On the other hand, classical approaches require $\mathcal{O}(NT)$ space for the state vector and adjoint vector, $\mathcal{O}(MN)$ space for coefficient matrices A_μ, and $\mathcal{O}(M)$ space for the design variables. Therefore, our algorithm

	Classical	Quantum
Gate complexity	$\tilde{\mathcal{O}}\left(\frac{\alpha_{\mathrm{PDE}}^{1+1/p} d^{1+\frac{1}{p}} N^{\frac{\eta}{d}(1+\frac{1}{p}+\frac{d}{\eta})} T^{1+\frac{1}{p}}}{\epsilon^{1/p}}\right)$	$\tilde{\mathcal{O}}\left(\frac{\alpha_{\mathrm{PDE}} d N^{\frac{\eta}{d}} M^3 T}{h_\xi^2 \epsilon}\right)$
Space complexity	$\mathcal{O}(NT + MN)$	$M\lceil \log_2(1/h_\xi + 1)\rceil + \tilde{\mathcal{O}}\left(\frac{\eta}{d}\log N + \log\left(\frac{\alpha_{\mathrm{PDE}} d M T}{h_\xi \epsilon}\right)\right)$

TABLE I. Gate and space complexities of the proposed method for strongly convex objective functions in the regime $N \gg M$ to achieve tolerance ε. For parameters regarding the underlying PDE, N is the number of degrees of freedom, d is the spatial dimension, T is the final time, α_{PDE} is the dominant coefficient, and η is the order of spatial derivative. For parameters regarding optimization, M is the number of design variables, and h_ξ is the interval of discretized design variables. p is the order of the time integrator in a classical PDE solver. The gate complexity for the classical algorithm corresponds to the number of arithmetic operators.

has an exponential space advantage over conventional classical approaches with respect to N and T.

5. Comparison of complexities

Here, we summarize the gate and space complexities of the proposed method compared with the conventional classical method, which are listed in Table I. As stated in Section III C 3, our algorithm has a better bound against the conventional classical algorithm, and has the potential for providing a polynomial speedup of degree $1 + d/\eta + 1/p$ for the system size of PDEs, N, an exponential speedup for the spatial dimension of PDEs, d, and a polynomial speedup of degree $1 + 1/p$ for the evolution time of PDEs, T. These speedups are attributed to those in the underlying PDE simulation. Also, as described in Section III C 4, our algorithm has an exponential space advantage over conventional classical approaches with respect to N and T. These results suggest that our algorithm is suitable for problems involving large-scale and high-dimensional simulations with a moderate number of parameters.

IV. APPLICATIONS

In this section, we provide two applications, one of which is a parameter calibration problem in the Black-Scholes equation, and the other of which is the material parameter design problem of the wave equation. In the following subsections, each problem and its motivation are first described, followed by numerical experiments. For numerical experiments, we used Python and its library Qiskit 1.3 [43], the open-source toolkit for quantum computation, to implement quantum circuits. We used the statevector simulation to validate the block-encoding, while we employed the direct calculation of the matrix-vector multiplication for the time-dependent Hamiltonian simulation. We employed a product formula for the time evolution of the time-dependent Hamiltonian for the simple implementation. We set the friction parameter ν, based on the average of the objective function $\nu = \sqrt{\bar{\mathcal{F}}}$, which is discussed in Appendix D.

A. Parameter calibration in Black-Scholes equation

1. Motivation and formulation

Here, we apply our framework to a parameter calibration problem in the financial option pricing problem. A derivative option is a financial asset whose future payoffs are determined by underlying assets, such as a stock price X. The present value of an option V is evaluated through mathematical models, and their model parameters must be learned from observed prices in financial markets. We demonstrate parameter calibration from a call option with the Black-Scholes model [44], which can be described by a linear parabolic PDE, formulated as

$$\frac{\partial V}{\partial \tau} + rX\frac{\partial V}{\partial X} + \frac{\sigma^2 X^2}{2}\frac{\partial^2 V}{\partial X^2} = rV, \tag{45}$$

where τ is time, r is a risk-free rate, and σ is the volatility, the standard deviation of the stock's returns. The final and boundary conditions are determined as

$$V(X,T) = \max(X - K, 0) \tag{46}$$

$$V(0,\tau) = 0 \tag{47}$$

$$\frac{\partial V(X,\tau)}{\partial X} = 1 \text{ as } X \to \infty, \tag{48}$$

where K is a strike price, and T is an expiration time of the option. By applying a change of variables $X = e^x$, $u = V - e^x$, $t = T - \tau$, we transform the Black-Scholes equation into the backward PDE with the initial and boundary conditions as follows:

$$\frac{\partial u}{\partial t} = \left(r - \frac{\sigma^2}{2}\right)\frac{\partial u}{\partial x} + \frac{\sigma^2}{2}\frac{\partial^2 u}{\partial x^2} - ru \tag{49}$$

$$u(x,0) = \max(e^x - K, 0) - e^x \tag{50}$$

$$u(x,t) = 0 \text{ as } x \to -\infty \tag{51}$$

$$\frac{\partial u(x,t)}{\partial x} = 0 \text{ as } x \to \infty, \tag{52}$$

which is a constant-coefficient PDE where t is the time until expiration. Here, we use the notation τ for a time and t for the time until expiration to make Eq. (49) consistent with our formulation in Eq. (2), although the symbols t and τ are conventionally defined in the opposite manner.

In this study, we consider an inverse problem to estimate the volatility σ when the call option price is observed in the financial market, i.e., the market price $u_{\mathrm{ref}} \approx u(x,T)$ is observed. To make the notation clear, we denote the model price by $u(x,T;\xi)$ instead of $u(x,T)$. The calibration problem is formulated as follows.

$$\begin{aligned} \min_{\xi\in[0,1]} \quad & \mathcal{F}_{\mathrm{err}}(\xi) = \|u(x,T;\xi) - u_{\mathrm{ref}}\|^2 \\ \text{subject to} \quad & \text{Eqs. (49)—(52)} \\ & \sigma^2 = (\sigma_{\max}^2 - \sigma_{\min}^2)\xi + \sigma_{\min}^2, \end{aligned} \tag{53}$$

where u_{ref} is the data point at $x = \log(X)$, and $\sigma_{\min}$ and $\sigma_{\max}$ are the lower and upper bounds of volatility, respectively. This forward problem is a simple yet well-studied example using quantum simulation [45, 46], while the analytical solution exists. We demonstrate the inverse problem having one variable σ since the other parameter r can typically be calibrated from other datasets independently. Also, (K,x,T) are not model parameters but variables to specify the date points. To solve this problem, we first truncate the domain to the interval $x \in [x_{\min}, x_{\max}]$ where $x_{\min}$ is sufficiently small and $x_{\max}$ is sufficiently large, and discretize it using a uniform grid with N nodes. Applying the finite difference method, the Black-Scholes equation reduces to an ODE, as follows.

$$\frac{\mathrm{d}u}{\mathrm{d}t} = \left(\left(r - \frac{\sigma^2}{2}\right) D_{\mathrm{D},N}^{\pm} + \frac{\sigma^2}{2} L_{\mathrm{D},N} - rI\right) u, \tag{54}$$

where $D_{\mathrm{D},N}^{\pm}$ and $L_{\mathrm{D},N}$ are central difference and discrete Laplace operators, respectively, both subject to Dirichlet boundary conditions on the left endpoint and Neumann conditions on the right endpoint. The coefficient matrix of this ODE is rearranged to the form in Eq. (3) as

$$\begin{aligned} A &= \left(r - \frac{\sigma^2}{2}\right) D_{\mathrm{D},N}^{\pm} + \frac{\sigma^2}{2} L_{\mathrm{D},N} - rI \\ &= \underbrace{\left(r - \frac{\sigma_{\min}^2}{2}\right) D_{\mathrm{D},N}^{\pm} + \frac{\sigma_{\min}^2}{2} L_{\mathrm{D},N} - rI}_{=A_0} \\ &\quad + \underbrace{\frac{\sigma_{\max}^2 - \sigma_{\min}^2}{2} (L_{\mathrm{D},N} - D_{\mathrm{D},N}^{\pm})}_{A_1} \xi. \end{aligned} \tag{55}$$

Thus, we need to implement the block-encoding of A_0 and A_1 to apply our framework. Based on the block-encoding of finite difference operators given in Appendix A, we can construct the block-encodings of A_0 and A_1 by $\mathcal{O}((\log N)^2)$ single and two-qubit gates. Alternatively, access to Hamiltonian simulation oracles $e^{-iH_0\tau}$, $e^{-iL_0\tau}$, $e^{-iH_1\tau}$ and $e^{-iL_1\tau}$ also suffices to apply our method, which we will employ in the following numerical experiments as the simplest implementation.

2. Numerical experiments

Here, we provide the results of numerical experiments to demonstrate the validity of our method. As parameters of the Black-Scholes equation, we set time to maturity $T = 1$, risk-free rate $r = 0.02$, and strike price $K = 30$. For truncating the domain, we set $x_{\min} = \ln(10^{-4})$ and $x_{\max} = \ln(10K)$, and the domain is discretized into a uniform grid of 64 nodes using 6 qubits. To implement the LCHS for forward simulation, we use Trotterization and the kernel $f(k) = 1/(\pi(1+ik))$ in Eq. (8) as the simplest implementation, where we use 8 qubits to discretize the truncated k-space from -64 (-2^6) to 63.5 $(2^6 - 2^{-1})$ with interval 0.5. We set the time increment of Trotterization to 0.02. For optimization, we discretize a parameter ξ using 4 qubits, and set the range of volatility as $\sigma_{\min} = 10^{-2}$ and $\sigma_{\max} = 0.5$. To prepare the reference state, i.e., the data points in Eq. (53), we use the analytical expression of the European call option price, which is given in the form [44, 46]:

$$V^*(X,\tau) = X\Phi(y_1) - Ke^{-r(T-\tau)}\Phi(y_2), \tag{56}$$

where

$$y_1 = \frac{\ln\frac{X}{K} + (r + \frac{\sigma^2}{2})(T-\tau)}{\sigma\sqrt{T-\tau}} \tag{57}$$

$$y_2 = y_1 - \sigma\sqrt{T-\tau}, \tag{58}$$

and $\Phi(y)$ is the cumulative distribution function of the standard normal distribution. Selecting a point in x-space, denoted by x_{ref}, which is the closest point to $\ln(Ke^{-r})$, we prepare a data point as

$$u_{\mathrm{ref}} = V^*(e^{x_{\mathrm{ref}}}, 0) - e^{x_{\mathrm{ref}}}, \tag{59}$$

where we choose $\sigma = 0.3$. The number of qubits required for block-encoding is 19, which consists of $n = 6$, $a_{\mathrm{for}} = 8$, $M = 1$, $m = 4$, and one additional ancillary qubit for LCU in Eq. (33).

Figure 4 illustrates the objective function values computed by the block-encoding and the forward simulation. In the case of forward simulation, we performed LCHS for each value of the design variable, and evaluated the objective function by the expectation value of the projector P. We confirmed that the objective function values computed by the block-encoding agree well with those by forward simulation, which demonstrates the validity of the proposed block-encoding. The optimal solution is $|\xi\rangle = |7\rangle$, which corresponds to $\xi = 7/15$.

Next, we performed optimization by the time-dependent Hamiltonian simulation to examine the validity of the proposed method. We empirically set the time increment $\Delta s = 0.01/\sqrt{\bar{\mathcal{F}}}$, and the total time $S = 10/\sqrt{\bar{\mathcal{F}}}$, which yields $N_{\mathrm{step}} = 1000$, where $\bar{\mathcal{F}}$ is the mean value of the objective function for all parameters, which can be evaluated by amplitude estimation (See Appendix D). Figure 5 shows the history of the expected

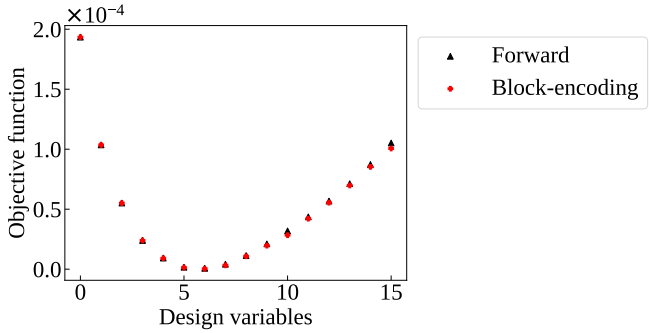

FIG. 4. The objective function values computed by the block-encoding and the forward simulation. The horizontal axis represents the index of the discretized design variables, i.e., decimal value of the basis $|\xi\rangle$.

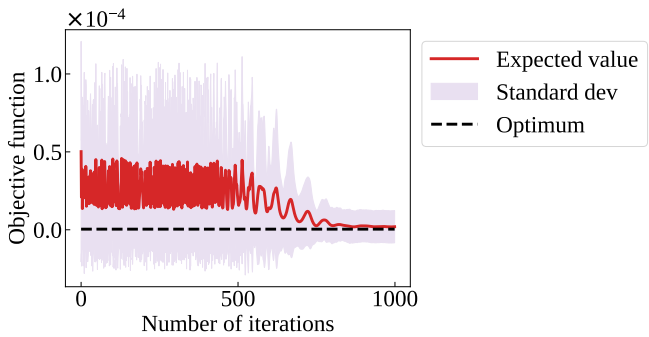

FIG. 5. The history of the expected value of the objective function (red line) with its standard deviation (purple region). The horizontal axis represents the number of iterations of the product formula. The black dashed lines represent the optimal value $\mathcal{F}(7/15)$.

value of the objective function $\langle\psi(t)|\,\hat{\mathcal{F}}\,|\psi(t)\rangle$ with its standard deviation $\sqrt{\langle\psi(t)|\,\hat{\mathcal{F}}^2\,|\psi(t)\rangle - \langle\psi(t)|\,\hat{\mathcal{F}}\,|\psi(t)\rangle^2}$. We observed that the expected value of the objective function converged to the optimum, while its variance decreased progressively. Figure 6 shows the distributions of all possible values of the design variable at the initial, intermediate, and final steps of the optimization. This figure clearly shows that the probability (blue distribution) concentrates on the optimal solution $\xi = 6/15$, which has the minimum value of the black plot.

We also examined the validity of the optimal solution. Figure 7 shows the forward simulation results for the optimal solution and the other reference solutions. The blue lines indicate option prices with respect to the stock price calculated by forward simulation. The black points represent data points used to define the reference state. In Figs. 7(a) and (b), the model fails to fit the data, whereas the optimized model in Fig. 7(c) shows good agreement with data points.

B. Material parameter design in wave equation

1. *Motivation and formulation*

We also apply our framework to a material parameter design problem of the one-dimensional wave equation. The wave equation describes the dynamics of waves such as vibrations of a string, sound, and light, appearing in various fields, including mechanics, acoustics, and electromagnetics. In wave-manipulating devices such as optical and acoustic devices, careful design for controlling reflection, transmission, and propagation velocity plays a critical role in determining device performance. Hence, we herein consider a material parameter design problem to concentrate the energy of waves in a certain region.

The one-dimensional wave equation is a linear hyperbolic PDE described as

$$\frac{\partial^2 w}{\partial t^2} = c(x)^2 \frac{\partial^2 w}{\partial x^2}, \tag{60}$$

where w is a state variable, and c is the speed of sound. We focus the equation on a segment $\Omega := (0, W)$ with the initial and boundary conditions

$$w(x,0) = w_0, \quad \frac{\partial w(x,0)}{\partial t} = v_0 \tag{61}$$

$$\frac{\partial w(0,t)}{\partial x} = 0, \quad w(W,t) = 0, \tag{62}$$

where w_0 and v_0 are the initial state and its time derivative, respectively. Since the speed of the wave depends on a material parameter in which the wave propagates, we parameterize it using design variables as follows.

$$c(x) = \begin{cases} c_1 & \text{if } x \le \frac{W}{4} \\ c_2 + (c_1 - c_2)\xi_1 & \text{if } \frac{W}{4} < x \le \frac{W}{2} \\ c_2 + (c_1 - c_2)\xi_2 & \text{if } \frac{W}{2} < x \le \frac{3W}{4} \\ c_2 & \text{otherwise,} \end{cases} \tag{63}$$

where c_1 is the speed of the wave in a material and c_2 is that in the other material. Then, we formulate the material parameter design problem as

$$\begin{aligned} \max_{\xi\in[0,1]^2} \quad & \int_{\frac{3W}{4}}^{1} \left(c(x)^2 w(x,T)^2 + \left(\frac{\partial w(x,T)}{\partial t}\right)^2 \right) \mathrm{d}x \\ \text{subject to} \quad & \text{Eqs. (60)—(63).} \end{aligned} \tag{64}$$

Here, we focus on the maximization problem, which can be converted into the minimization problem by adding the minus sign to the objective function. This corresponds to the replacement of the reflection operator R_P into $-R_P$ in constructing the block-encoding of the objective function. Figure 8 illustrates the problem setting of the material parameter design. The impulse wave is provided at the left boundary and the optimization aims to maximize the wave energy at the gray region by designing speed of wave (material parameters) of the intermediate domains. To solve this problem, we discretize

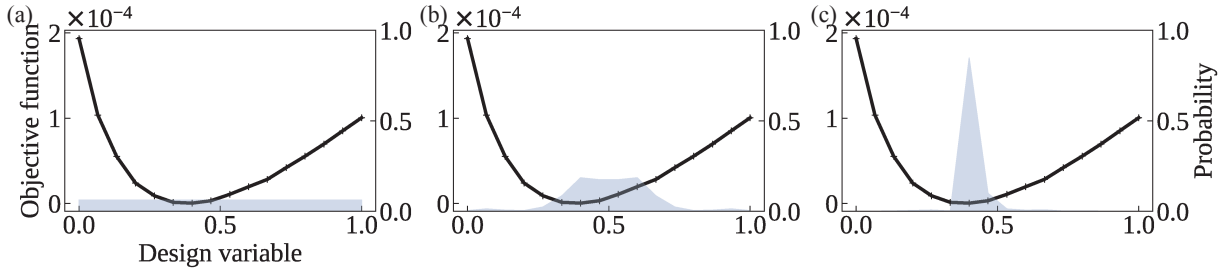

FIG. 6. The probability distribution of all possible design variables at (a) initial, (b) intermediate (100th), and (c) final steps of the optimization. The black line shows the landscape of the objective function, and the blue region represents the probability distribution of the design variable.

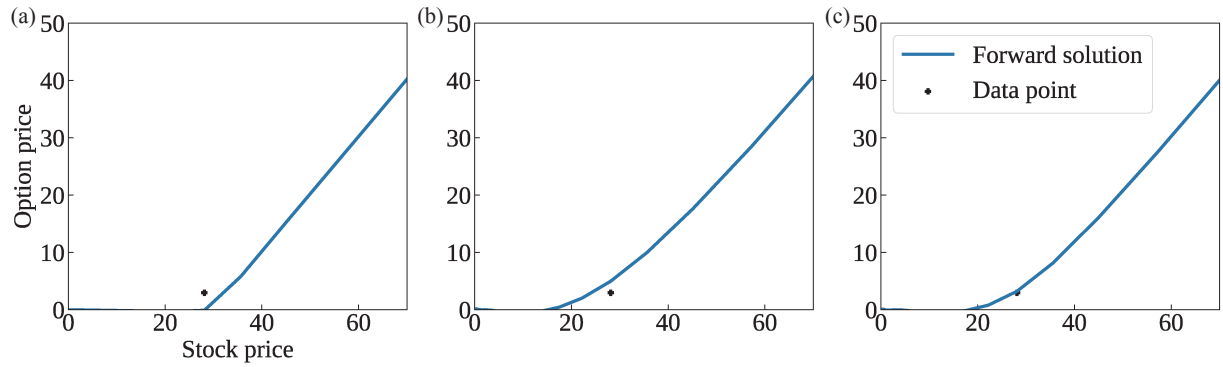

FIG. 7. The forward simulation results for design variables (a) $\xi = 0$, (b) $\xi = 1$, and (c) $\xi = 6/15$. Blue lines illustrate the plots of the option price vs stock price obtained by forward simulation. The black points are data points used to define the reference state.

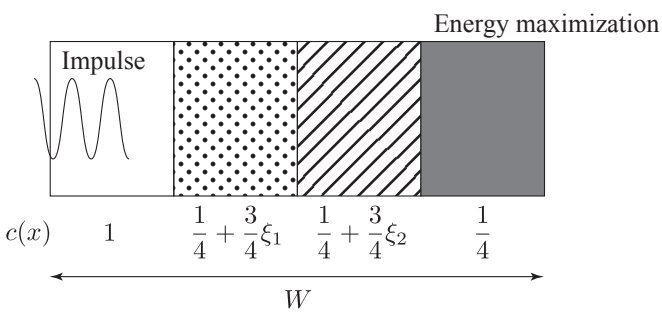

FIG. 8. Problem setting of the material parameter design in the wave equation.

the wave equation in the direction of space and rearrange it to the following form:

$$\frac{\mathrm{d}}{\mathrm{d}t}\underbrace{\begin{pmatrix} w^{\mathrm{t}} \\ w^{\mathrm{x}} \end{pmatrix}}_{u} = -i\underbrace{\begin{pmatrix} 0 & CD_{\mathrm{D}}^{-} \\ D_{\mathrm{D}}^{+}C & 0 \end{pmatrix}}_{A}\underbrace{\begin{pmatrix} w^{\mathrm{t}} \\ w^{\mathrm{x}} \end{pmatrix}}_{u}, \tag{65}$$

where w^{t} and w^{x} are discretized state vectors of $(1/c(x))\partial w/\partial t$ and $\partial w/\partial x$, respectively, D_{D}^{+} and D_{D}^{-} are forward and backward difference operators with the Dirichlet boundary conditons, respectively. The matrix C is a diagonal matrix whose j-th entry corresponds to the speed of wave $c(x)$ at the j-th node. That is, based on the parameterization in Eq. (63), it can be decomposed into the following form:

$$C = c_2 I + C_0 + C_1\xi_1 + C_2\xi_2, \tag{66}$$

where C_0, C_1, and C_2 are diagonal matrices whose entries are $c_1 - c_2$ for nodes satisfying $x \leq W/4$, $W/4 < x \leq W/2$, and $W/2 < x \leq 3W/4$, respectively, and 0 otherwise. The coefficient matrix of this ODE is then rearranged to the form in Eq. (3) as

$$\begin{aligned} A &= i\begin{pmatrix} 0 & CD_{\mathrm{D}}^{-} \\ D_{\mathrm{D}}^{+}C & 0 \end{pmatrix} \\ &= \underbrace{i\begin{pmatrix} 0 & \left(\frac{1}{4}I + C\right)D_{\mathrm{D}}^{-} \\ D_{\mathrm{D}}^{+}\left(\frac{1}{4}I + C\right) & 0 \end{pmatrix}}_{A_0} \\ &\quad + \underbrace{i\begin{pmatrix} 0 & C_1 D_{\mathrm{D}}^{-} \\ D_{\mathrm{D}}^{+}C_1 & 0 \end{pmatrix}}_{A_1}\xi_1 + \underbrace{i\begin{pmatrix} 0 & C_2 D_{\mathrm{D}}^{-} \\ D_{\mathrm{D}}^{+}C_2 & 0 \end{pmatrix}}_{A_2}\xi_2. \end{aligned} \tag{67}$$

Thus, we need to implement the block-encodings of A_0, A_1, and A_2 to apply our framework. Alternatively, it is

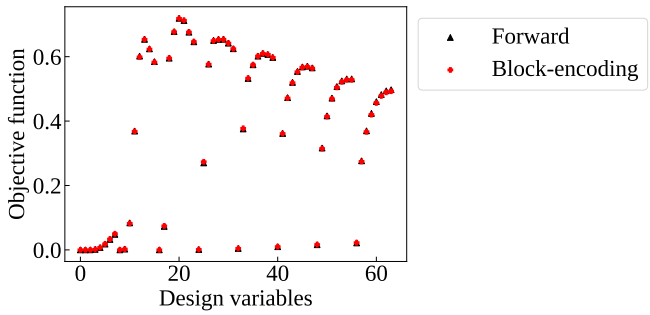

FIG. 9. The objective function values evaluated by the block-encoding and the forward simulation. The horizontal axis represents the index of the discretized design variables, i.e., decimal value of the basis $|\xi_2, \xi_1\rangle$.

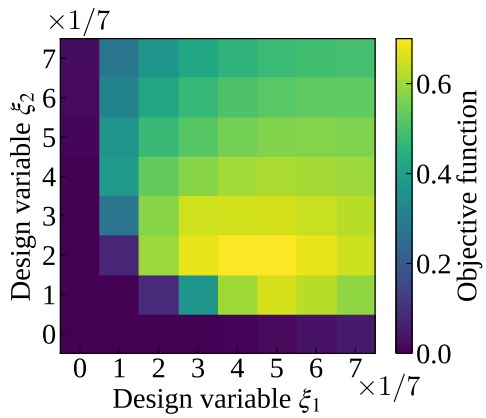

FIG. 10. The objective function values evaluated by the block-encoding. Each axis represents each index of the discretized design variables, i.e., decimal value of the basis $|\xi_1\rangle$ and $|\xi_2\rangle$, respectively.

also sufficient to implement Hamiltonian simulation oracles $e^{-iH_0\tau}$, $e^{-iH_1\tau}$, and $e^{-iH_2\tau}$, which we will use in the subsequent experiments as the simplest implementation. Note that since A_0, A_1, and A_2 are anti-Hermitian, we perform simple Hamiltonian simulation, rather than LCHS.

2. *Numerical experiments*

Here, we provide numerical examples to demonstrate the validity of our method. As parameters of the wave equation, we set the domain length $W = 2^5 - 1$, simulation time $T = 48$, and the speed of the wave $c_1 = 1$ and $c_2 = 1/4$. We set the initial conditions as

$$w_0 = 0 \tag{68}$$

$$v_0 = \begin{cases} e^{-\frac{x^2}{2\cdot 2^2}} & x < 4 \\ 0 & x \geq 4, \end{cases} \tag{69}$$

which is sparse, and thus can be prepared efficiently. We implement the state preparation oracle by amplitude encoding [47] through Qiskit. We discretize the segment into a uniform grid using 5 qubits with one additional ancillary qubit to encode the wave equation into an ODE in Eq. (65), which corresponds to $n = 6$. For forward simulation, we use Trotterization with the time increment 0.5. For optimization, we discretize each parameter ξ_1 ξ_2 using 3 qubits. The number of qubits required for block-encoding is 12, which consists of $n = 6$, $a_{\text{for}} = 0$, $M = 2$, $m = 3$.

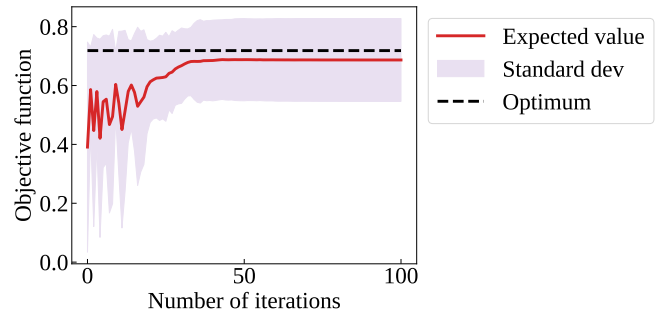

FIG. 11. The history of the expected value of the objective function (red line) with the purple region indicating the standard deviation of the objective function. The horizontal axis represents the number of iterations of the product formula. The red lines illustrate the expected value of the objective function, while the black dashed lines represent the optimal value $\mathcal{F}(4/7, 2/7)$.

Figure 9 illustrates the objective function values evaluated by the block-encoding and the forward simulation. In forward simulation, we evaluated the objective function for each value of the design variable by performing Hamiltonian simulation, followed by computing the expectation value of the projector P. As shown in Fig. 9, the objective function values computed by the block-encoding agree well with those obtained by forward simulation. Figure 10 shows the landscape of the objective function computed by its block-encoding. Since we focus on the maximization problem, the optimal solution is $|\xi_2, \xi_1\rangle = |2, 4\rangle$, which corresponds to $\xi_1 = 4/7$ and $\xi_2 = 2/7$.

Next, we provide numerical experiments of optimization by the time-dependent Hamiltonian simulation to verify whether our method can find the optimal solution. We set the time increment $\Delta s = 0.1/\sqrt{\bar{\mathcal{F}}}$, and the total time $S = 10/\sqrt{\bar{\mathcal{F}}}$, which yields $N_{\text{step}} = 100$. Figure 11 shows the history of the expected value of the objective function $\langle\psi(t)|\,\hat{\mathcal{F}}\,|\psi(t)\rangle$ with its standard deviation $\sqrt{\langle\psi(t)|\,\hat{\mathcal{F}}^2\,|\psi(t)\rangle - \langle\psi(t)|\,\hat{\mathcal{F}}\,|\psi(t)\rangle^2}$, where $\hat{\mathcal{F}} := (U_{\text{obj}} + I)/2$ is an objective function operator. The expected value of the objective function converged near the optimal value with a gradual decrease in the standard deviation. As the standard deviation appears to remain moderately large, we next examined the probability distributions of the design variables. Figure 12 shows the distributions of the design variables at the initial, inter-

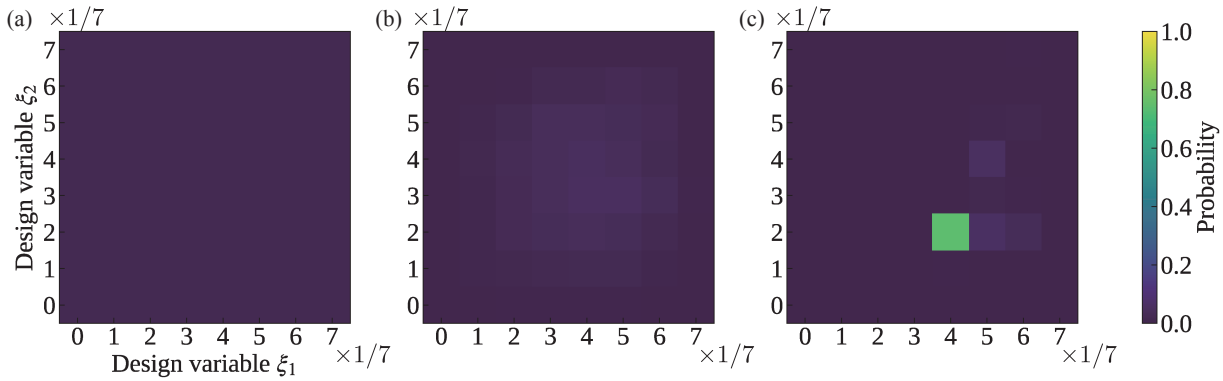

FIG. 12. The probability distribution of all possible design variables at (a) initial, (b) intermediate (10th), and (c) final steps of the optimization.

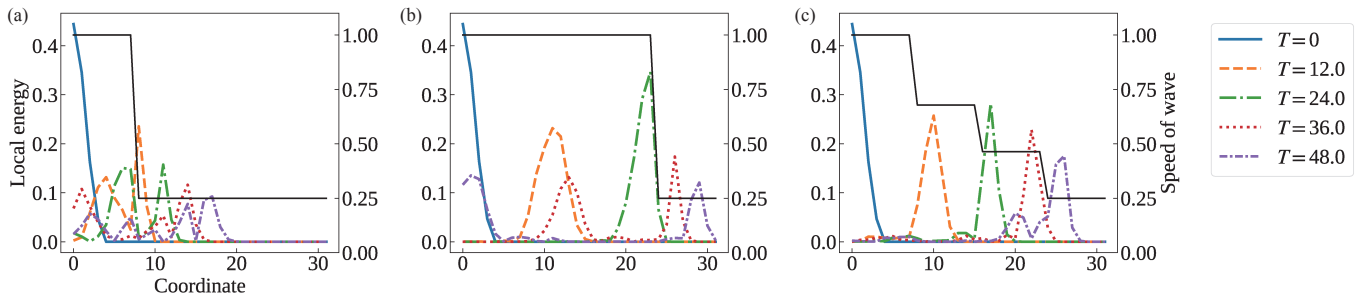

FIG. 13. The forward simulation results for design variables (a) $(\xi_1, \xi_2) = (0, 0)$, (b) $(\xi_1, \xi_2) = (1, 1)$, and (c) $(\xi_1, \xi_2) = (4/7, 2/7)$. Each color line illustrates the local energy of the wave field at each time slice. The black line shows the speed of wave at each spatial coordinate, which depends on the design variables.

mediate, and final steps of the optimization. This figure clearly shows that the probability concentrates on the optimal solution $(\xi_1, \xi_2) = (4/7, 2/7)$.

Finally, we examined the validity of the optimal solution. Figure 13 illustrates the forward simulation results for the optimal solution and the other reference solutions. Each color line illustrates the local energy of the wave field at each time slice, while the black line shows the speed of wave at each spatial coordinate, depending on the design variables. In Figs. 13(a) and (b), the wave reflects at the boundary of two phases (areas with $c(x) = 1$ and $c(x) = 1/4$), which prevents the wave from propagating smoothly into the forward direction. In the optimal solution in Fig. 13, on the other hand, the speed of wave changes in a stepwise manner, which suppresses the reflection at the boundary of adjacent phases. This resulting structure has typical characteristics of antireflection (AR) designs; its discovery through optimization validates the proposed method.

V. CONCLUSIONS

This study proposed a quantum algorithm for PDE-constrained optimization with continuous design variables. Specifically, we constructed an explicit block-encoding of the objective function that coherently bridges a quantum PDE solver and a quantum optimizer, thereby avoiding any classical readout of the output of the quantum PDE solver, which could negate the potential quantum speedups. We then derived the query and gate complexities of the proposed method. Under the assumption that the optimization landscape is strongly convex, we showed that, compared with conventional classical approaches, our algorithm can provide a polynomial speedup for the system size of PDEs N, an exponential speedup for the spatial dimension of PDEs, d, and a polynomial speedup for the evolution time of PDEs T, in the regime where the system size of discretized PDEs is much larger than the number of design variables. Our algorithm also exhibited the exponential space advantage over classical approaches with respect to the system size of PDEs. These results suggest that quantum optimization can inherit quantum speedup from the underlying quantum PDE solver. As numerical experiments, we demonstrated two applications: a parameter calibration problem in the Black-Scholes equation and a material parameter design problem in the wave equation, which support the effectiveness of the proposed method. More broadly, this work presents the concept of composing quantum subroutines so that the prohibitive readout overhead of one can be avoided by using it coherently as

input to another, paving the way to exploiting quantum speedups in practical scenarios.

ACKNOWLEDGEMENT

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Appendix A: Block-encoding of finite difference operator

Let us consider a one-dimensional domain $\Omega = (0, 1)$ in which a scalar field $u(x)$ is defined with x the spatial coordinate. For finite difference approximation, we discretize the domain Ω into a grid with N_x nodes, and represent the scalar field by its value at each node, i.e., $u := [u_1, \ldots, u_{N_x}]^\top$ where u_j is the value at the j-th node. The forward, central, and backward difference schemes, denoted by D^+, D^-, and $D^\pm$ respectively, approximate the first-order spatial derivative as follows:

$$\begin{aligned}(D^+u)_j &= \frac{u_{j+1} - u_j}{h} + O(h),\\ (D^-u)_j &= \frac{u_j - u_{j-1}}{h} + O(h),\\ (D^\pm u)_j &= \frac{u_{j+1} - u_{j-1}}{2h} + O(h^2),\end{aligned} \tag{A1}$$

where h is the interval of adjacent nodes. Similarly, the Laplace operator by the central difference scheme is given as

$$(\Lambda u)_j = \frac{u_{j+1} - 2u_j + u_{j-1}}{h^2} + O(h^2). \tag{A2}$$

With the Dirichlet boundary condition, these finite difference operators can be represented by using the identity and shift operators as follows:

$$\begin{aligned}D^+ &= \frac{1}{h}\left(\texttt{SHIFT}^+ - I - \sigma_{10}^{\otimes n_x}\right)\\ D^- &= \frac{1}{h}\left(I - \texttt{SHIFT}^- + \sigma_{01}^{\otimes n_x}\right)\\ D^\pm &= \frac{1}{2h}\left(\texttt{SHIFT}^+ - \texttt{SHIFT}^- + \sigma_{01}^{\otimes n_x} - \sigma_{10}^{\otimes n_x}\right)\\ \Lambda &= \frac{1}{h^2}\left(\texttt{SHIFT}^+ + \texttt{SHIFT}^- - 2I - \sigma_{01}^{\otimes n_x} - \sigma_{10}^{\otimes n_x}\right)\end{aligned} \tag{A3}$$

where $n_x := \log_2 N_x$, $\texttt{SHIFT}^+ := \sum_{j=0}^{2^{n_x}-1} |j+1\rangle\langle j|$, $\texttt{SHIFT}^- := \sum_{j=0}^{2^{n_x}-1} |j\rangle\langle j+1|$, $\sigma_{01} := |0\rangle\langle 1|$, and $\sigma_{10} := |1\rangle\langle 0|$. The terms including ladder operators σ_{01} and σ_{10} are for the Dirichlet boundary conditions. Here, we construct a block-encoding of the following Hermitian operator $B(\alpha_1, \alpha_2, \phi)$:

$$\begin{aligned}B(\alpha_1, \alpha_2, \phi) := {}& \alpha_1 e^{i\phi}(\texttt{SHIFT}^+ - \sigma_{10}^{\otimes n_x})\\ &+ \alpha_1 e^{-i\phi}(\texttt{SHIFT}^- - \sigma_{01}^{\otimes n_x}) + \alpha_2 I,\end{aligned} \tag{A4}$$

where α_1, α_2 are real coefficients, and λ is a phase parameter. For example, $B(1/h^2, -2/h^2, 0)$ corresponds to the Laplace operator while $B(1/2h, 0, \pi/2)$ to the central difference operator up to global phase $e^{i\pi/2}$. Since the terms of ladder operators are not unitary, we rearrange them as follows.

$$\begin{aligned}&e^{-i\phi}\sigma_{01}^{\otimes n_x} + e^{i\phi}\sigma_{10}^{\otimes n_x}\\ &= \frac{|0\rangle^{\otimes n_x} + e^{i\phi}|1\rangle^{\otimes n_x}}{\sqrt{2}}\frac{\langle 0|^{\otimes n_x} + e^{-i\phi}\langle 1|^{\otimes n_x}}{\sqrt{2}}\\ &\quad - \frac{|0\rangle^{\otimes n_x} - e^{i\phi}|1\rangle^{\otimes n_x}}{\sqrt{2}}\frac{\langle 0|^{\otimes n_x} - e^{-i\phi}\langle 1|^{\otimes n_x}}{\sqrt{2}}\\ &= U_{\mathrm{GHZ}}(\phi) Z \otimes |0\rangle\langle 0|^{\otimes n_x - 1} U_{\mathrm{GHZ}}(\phi)^\dagger,\\ &= \frac{1}{2}U_{\mathrm{GHZ}}(\phi)(Z \otimes R_0)U_{\mathrm{GHZ}}(\phi)^\dagger\\ &\quad + \frac{1}{2}U_{\mathrm{GHZ}}(\phi)(Z \otimes I)U_{\mathrm{GHZ}}(\phi)^\dagger,\end{aligned} \tag{A5}$$

where Z is a Pauli Z operator, $R_0 := 2|0\rangle\langle 0|^{\otimes n_x - 1} - I$ is a reflection operator, and $U_{\mathrm{GHZ}}(\phi)$ is a unitary that prepares $U_{\mathrm{GHZ}}|0\rangle^{\otimes n_x} = (|0\rangle^{\otimes n_x} + e^{i\phi}|1\rangle^{\otimes n_x})/\sqrt{2}$ and $U_{\mathrm{GHZ}}|1\rangle|0\rangle^{\otimes n_x - 1} = (|0\rangle^{\otimes n_x} - e^{i\phi}|1\rangle^{\otimes n_x})/\sqrt{2}$. Then, the Hermitian operator B is written as a linear combination of unitaries with five terms, as follows:

$$\begin{aligned}B(\alpha_1, \alpha_2, \phi) = {}& \alpha_1 e^{i\phi}\texttt{SHIFT}^+ + \alpha_1 e^{-i\phi}\texttt{SHIFT}^- + \alpha_2 I\\ &- \frac{\alpha_1}{2}U_{\mathrm{GHZ}}(\phi)(Z \otimes R_0)U_{\mathrm{GHZ}}(\phi)^\dagger\\ &- \frac{\alpha_1}{2}U_{\mathrm{GHZ}}(\phi)(Z \otimes I)U_{\mathrm{GHZ}}(\phi)^\dagger,\end{aligned} \tag{A6}$$

which means that LCU yields $(3|\alpha_1| + |\alpha_2|, 3, 0)$-block-encoding of B. The shift operators include j-controlled Toffoli gates for $j = 0, 1, \ldots, n_x - 1$, which requires $\mathcal{O}(j)$ single and two-qubit gates [48]. Thus, the shift operators require $\mathcal{O}(n_x^2)$ single and two-qubit gates. On the other hand, the fourth and fifth terms yield $\mathcal{O}(n_x)$ single and two-qubit gates. Consequently, the number of single and two-qubit gates for implementing the block-encoding of B is $\mathcal{O}(n_x^2)$.

For d-dimensional cases, the Hermitian operator is extended to

$$\begin{aligned}&B^{(d)}\left(\{\alpha_1^{(\mu)}\}_{\mu=1}^d, \{\alpha_2^{(\mu)}\}_{\mu=1}^d, \{\phi^{(\mu)}\}_{\mu=1}^d\right)\\ &= \sum_{\mu=1}^d I^{\otimes(d-\mu)} \otimes B(\alpha_1^{(\mu)}, \alpha_2^{(\mu)}, \phi^{(\mu)}) \otimes I^{\otimes(\mu-1)},\end{aligned} \tag{A7}$$

which is also the form of a linear combination of unitaries with $5d$ terms. This means that LCU

yields $(3\sum_{\mu=1}^{d}|\alpha_1^{(\mu)}| + \sum_{\mu=1}^{d}|\alpha_2^{(\mu)}|, \lceil\log(5d)\rceil, 0)$-block-encoding of $B^{(d)}$, which requires $\mathcal{O}(n^2/d)$ single and two-qubit gates where we assume that we assign n/d qubits for each dimension. For example, an operator $\alpha_{\rm PDE}\nabla^\eta$ in the d-dimensional space with the order of spatial derivative η can be discretized by $B^{(d)}$ by substituting $\alpha_1^{(\mu)} = \alpha_{\rm PDE}/h_x^\eta$, $\alpha_2^{(\mu)} = -2\alpha_{\rm PDE}/h_x^\eta$, and $\phi^{(\mu)} = 0$.

Since $B^{(d)}\left(\{\alpha_1^{(\mu)}\}_{\mu=1}^d, \{\alpha_2^{(\mu)}\}_{\mu=1}^d, \{\phi^{(\mu)}\}_{\mu=1}^d\right)$ corresponds to the discrete Laplace operator $\Lambda^{(M)}$ when $\alpha_1^{(\mu)} = 1/h_\xi^2$, $\alpha_2^{(\mu)} = -2/h_\xi^2$, $\phi^{(\mu)} = 0$, and $d = M$, $n = mM$, we can construct the $(5M/h_\xi^2, \lceil\log(5M)\rceil, 0)$-block-encoding of $\Lambda^{(M)}$ by using $\mathcal{O}(Mm^2)$ single and two-qubit gates.

Appendix B: Block-encoding of spatially varying coefficients of PDEs

Here, we construct a block-encoding of the diagonal matrix derived from the spatially varying coefficients of PDEs. Let $c(x)$ denote a spatially varying coefficient of a PDE such as speed of sound for acoustic simulation. Discretizing $c(x)$ by the finite difference method yields a diagonal matrix C whose (j,j)-entry corresponds to the value of $c(x)$ at the j-th node of the spatial grid. In this study, we assume that $c(x)$ is a piece-wise constant function and the corresponding diagonal matrix C can be represented using projectors $\sigma_{00} := |0\rangle\langle 0|$ and $\sigma_{11} := |1\rangle\langle 1|$ for a constant value N_C, as follows:

$$C = \sum_{k=1}^{N_{\rm C}} \alpha_{\rm PDE}^{(k)} C_n^{(k)} \otimes \cdots \otimes C_1^{(k)}, \tag{B1}$$

where $C_j^{(k)} \in \{\sigma_{00}, \sigma_{11}, I\}$, and $\alpha_{\rm PDE}^{(k)}$ is a coefficient. Introducing a reflection operator $R_C^{(k)} := 2C_n^{(k)}\otimes\cdots\otimes C_1^{(k)} - I$, we can represent the diagonal matrix as

$$C = \sum_{k=1}^{N_{\rm C}} \alpha_{\rm PDE}^{(k)} \frac{R_C^{(k)} + I}{2} = \sum_{k=1}^{N_{\rm C}} \frac{\alpha_{\rm PDE}^{(k)}}{2} R_C^{(k)} + \frac{I}{2}\sum_{k=1}^{N_{\rm C}} \alpha_{\rm PDE}^{(k)}, \tag{B2}$$

which is LCU requiring $\mathcal{O}(nN_c)$ single and two-qubit gates since each reflection operator requires Z gates controlled by at most $n-1$-qubits. The operator norm of C is bounded by $\alpha_{\rm PDE} := N_C \max_k \alpha_{\rm PDE}^{(k)}$. Thus, we obtain the $(\sum_k |\alpha_{\rm PDE}^{(k)}|, \lceil\log(N_C)\rceil, 0)$-block-encoding of C. Note that we assume $N_c \in \mathcal{O}(1)$ in the complexity analysis in Section III C 1.

Appendix C: Block-encoding of the time-dependent Hamiltonian

Here, we derive the block-encoding of the time-dependent Hamiltonian in the interaction picture, $\mathcal{H}_{\rm I}(t)$, in Eq. (18) to simulate the time evolution

$$U_{\rm TDS}(s_{j+1}, s_j) := \mathcal{T}e^{-i\int_{s_j}^{s_{j+1}} \mathcal{H}_{\rm I}(\tau){\rm d}\tau}. \tag{C1}$$

Specifically, based on Ref. [35], we implement the block-encoding $\texttt{HAM-T}_j$ such that

$$(\langle 0|^{\otimes a_{\rm HT}} \otimes I^{\otimes(Mm+\lceil\log(Q)\rceil)})\texttt{HAM-T}_j(|0\rangle^{\otimes a_{\rm HT}} \otimes I^{\otimes(Mm+\lceil\log(Q)\rceil)}) = \sum_{q=0}^{Q-1} |q\rangle\langle q| \otimes \frac{1}{\alpha_\Lambda}\mathcal{H}_{\rm I}\left(s_j + \frac{q\Delta s}{Q}\right), \tag{C2}$$

where $Q \in \mathcal{O}(\frac{S}{\epsilon}(\alpha_\Lambda + \alpha_{\rm obj}e^{\nu S} + \nu))$ is the number of quadrature points for time integration from $s_j(= j\Delta s)$ to $s_{j+1}(= (j+1)\Delta s)$, $a_{\rm HT}$ is the number of ancillary qubits, and α_Λ is a normalization constant for block-encoding. According to Eq. (18), the above equation (C2) can be further decomposed into:

$$\sum_{q=0}^{Q-1} |q\rangle\langle q| \otimes \frac{1}{\alpha_\Lambda}\mathcal{H}_{\rm I}(t_{j,q}) = \underbrace{\left(\sum_{q=0}^{Q-1} |q\rangle\langle q| \otimes e^{i\frac{1}{\nu}(e^{\nu t_{j,q}}-1)\hat{\mathcal{F}}}\right)}_{=\widetilde{\mathcal{F}}(t_{j,q})^\dagger} \underbrace{\left(\sum_{q=0}^{Q-1} |q\rangle\langle q| \otimes e^{-\nu t_{j,q}}\frac{\Lambda^{(M)}}{2\alpha_\Lambda}\right)}_{=:\widetilde{\Lambda}(t_{j,q})} \underbrace{\left(\sum_{q=0}^{Q-1} |q\rangle\langle q| \otimes e^{-i\frac{1}{\nu}(e^{\nu t_{j,q}}-1)\hat{\mathcal{F}}}\right)}_{=:\widetilde{\mathcal{F}}(t_{j,q})} \tag{C3}$$

where $t_{j,q} := s_j + \frac{q\Delta s}{Q}$. Thus, we focus on the implementation of the block-encoding of $\widetilde{\mathcal{F}}(t_{j,q})$ and $\widetilde{\Lambda}(t_{j,q})$.

The implementation of the block-encoding of $\widetilde{\mathcal{F}}(t_{j,q})$ consists of the following steps:

1. Encode binary representation of $\frac{1}{\nu}(e^{\nu t_{j,q}}-1)$ into $\lceil\log\big(e^{\nu S}/(\nu\epsilon')\big)\rceil$ ancillary qubits within an error ϵ', i.e.,

$$U_{\mathrm{EN}}\left|q\right\rangle\left|0\right\rangle^{\otimes\lceil\log\left(e^{\nu S}/(\nu\epsilon')\right)\rceil} = \left|q\right\rangle\left|\frac{1}{\nu}(e^{\nu t_{j,q}}-1)\right\rangle, \tag{C4}$$

   which can be implemented efficiently using $\mathcal{O}(\lceil\log\big(e^{\nu S}/(\nu\epsilon')\big)\rceil^2)$ Toffoli gates [49].

2. Perform the time evolution of the objective function controlled by each ancillary qubit as

$$\prod_{b=1}^{\lceil\log\left(e^{\nu S}/(\nu\epsilon')\right)\rceil} C_b\left(e^{-i2^{b-1}\epsilon'\hat{\mathcal{F}}}\right), \tag{C5}$$

   where $C_b(e^{-i2^{b-1}\epsilon'\hat{\mathcal{F}}})$ represents the operation $e^{-i2^{b-1}\epsilon'\hat{\mathcal{F}}}$ controlled by the b-th ancillary qubit. Combining QSVT with OAA [42, Corollary 62], we can implement $(1, a_{\mathrm{obj}}+2, \epsilon'')$-block-encoding of $C_b\left(e^{-i2^{b-1}\epsilon'\hat{\mathcal{F}}}\right)$ using $6\alpha_{\mathrm{obj}}2^{b-1}\epsilon''+9\log(12/\epsilon'')$ queries to $(\alpha_{\mathrm{obj}}, a_{\mathrm{obj}}, \epsilon''/(2^{b-1}\epsilon')$-block-encoding of the objective function $\hat{\mathcal{F}}$.

By setting $\epsilon'' < \epsilon'/\lceil\log\big(e^{\nu S}/(\nu\epsilon')\big)\rceil$ so that the error of implementing Eq. (C5) could be within ϵ', we can construct $(1, a_{\mathrm{obj}}+2+\lceil\log\big(e^{\nu S}/(\nu\epsilon')\big)\rceil, 2\epsilon')$-block-encoding of $\widetilde{\mathcal{F}}(t_{j,q})$ using $\mathcal{O}(\alpha_{\mathrm{obj}}e^{\nu S}/\nu + \log\big(e^{\nu S}/\nu\epsilon'\big)\log\big(\log\big(e^{\nu S}/\nu\epsilon'\big)/\epsilon'\big))$ queries to $(\alpha_{\mathrm{obj}}, a_{\mathrm{obj}}, \nu\epsilon'/(e^{\nu S}\log\big(e^{\nu S}/\nu\epsilon'\big))$-block-encoding of the objective function $\hat{\mathcal{F}}$ or its inverse. Similarly, we can implement the block-encoding of $\widetilde{\mathcal{F}}(t_{j,q})^\dagger$. Given that the gate complexity of the $(\alpha_{\mathrm{obj}}, a_{\mathrm{obj}}, \epsilon''')$-block-encoding of the objective function $\hat{\mathcal{F}}$ for an error ϵ''' is $\tilde{\mathcal{O}}\left(\alpha_{\mathrm{PDE}}dN^{\frac{\eta}{d}}M^2T\log(\alpha_{\mathrm{for}}/\epsilon''')^{1/\beta}\right)$ based on the discussion in Section III C 1, the gate complexity of $(1, a_{\mathrm{obj}}+2+\lceil\log\big(e^{\nu S}/(\nu\epsilon')\big)\rceil, 2\epsilon')$-block-encoding of $\widetilde{\mathcal{F}}(t_{j,q})$ results in

$$\tilde{\mathcal{O}}\left(\frac{1}{\nu}e^{\nu S}\alpha_{\mathrm{PDE}}dN^{\frac{\eta}{d}}M^2T\log\left(\frac{e^{\nu S}}{\nu\epsilon'}\right)^{2+\frac{1}{\beta}}\right), \tag{C6}$$

where we regard $\alpha_{\mathrm{obj}}\in\mathcal{O}(1)$.

To implement the block-encoding of $\widetilde{\Lambda}(t_{j,q})$, we rearrange the terms, as follows:

$$\begin{aligned}\widetilde{\Lambda}(t_{j,q}) &= \sum_{q=0}^{Q-1}\left|q\right\rangle\!\left\langle q\right|\otimes e^{-\nu t_{j,q}}\frac{\Lambda^{(M)}}{2\alpha_\Lambda}\\ &= \left(\sum_{q=0}^{Q-1}e^{-\nu t_{j,q}}\left|q\right\rangle\!\left\langle q\right|\right)\otimes\frac{\Lambda^{(M)}}{2\alpha_\Lambda},\end{aligned} \tag{C7}$$

which means that $\widetilde{\Lambda}(t_{j,q})$ is separable. Since the discussion in Appendix A provides a block-encoding of $\Lambda^{(M)}$, it suffices to consider the implementation of the block-encoding of $\sum_{q=0}^{Q-1}e^{-\nu t_{j,q}}\left|q\right\rangle\!\left\langle q\right|$. To this end, we further rearrange the term, as follows:

$$\begin{aligned}\sum_{q=0}^{Q-1}e^{-\nu t_{j,q}}\left|q\right\rangle\!\left\langle q\right| &= e^{-\nu j\Delta s}\sum_{q=0}^{Q-1}(e^{-\nu\frac{\Delta s}{Q}})^q\left|q\right\rangle\!\left\langle q\right|\\ &= e^{-\nu j\Delta s}\sum_{q_{\lceil\log Q\rceil}=0}^{1}\cdots\sum_{q_1=0}^{1}(e^{-\nu\frac{\Delta s}{Q}})^{\sum_{b=1}^{\lceil\log Q\rceil}2^{b-1}q_b}\left|q_{\lceil\log Q\rceil},\ldots,q_1\right\rangle\!\left\langle q_{\lceil\log Q\rceil},\ldots,q_1\right|\\ &= e^{-\nu j\Delta s}\sum_{q_{\lceil\log Q\rceil}=0}^{1}(e^{-\nu\frac{\Delta s}{Q}})^{2^{\lceil\log Q\rceil-1}q_1}\left|q_{\lceil\log Q\rceil}\right\rangle\!\left\langle q_{\lceil\log Q\rceil}\right|\otimes\cdots\otimes\sum_{q_1=0}^{1}(e^{-\nu\frac{\Delta s}{Q}})^{q_1}\left|q_1\right\rangle\!\left\langle q_1\right|,\end{aligned} \tag{C8}$$

which means that this term is also separable. Each local operator can be represented as LCU, as follows.

$$\begin{aligned}&\sum_{q_b=0}^{1}(e^{-\nu\frac{\Delta s}{Q}})^{2^{b-1}q_b}\left|q_b\right\rangle\!\left\langle q_b\right|\\ &= (e^{-\nu\frac{\Delta s}{Q}})^{2^{b-1}}\left|1\right\rangle\!\left\langle 1\right|+\left|0\right\rangle\!\left\langle 0\right|\\ &= \frac{1+(e^{-\nu\frac{\Delta s}{Q}})^{2^{b-1}}}{2}I+\frac{1-(e^{-\nu\frac{\Delta s}{Q}})^{2^{b-1}}}{2}Z,\end{aligned} \tag{C9}$$

where the ℓ_1-norm of the coefficients is one. That is, LCU yields the $(1,1,0)$-block-encoding of each local operator, which means that we can implement the $(1,\lceil\log(Q)\rceil,0)$-block-encoding of $\sum_{q=0}^{Q-1} e^{-\nu t_{j,q}} |q\rangle\langle q|$. Combined with $(5M/h_\xi^2,\lceil\log(5M)\rceil,0)$-block-encoding of $\Lambda^{(M)}$, we obtain $(5M/h_\xi^2,\lceil\log(5M)\rceil+\lceil\log(Q)\rceil,0)$-block-encoding of $\widetilde{\Lambda}(t_{j,q})$, which uses

$$\mathcal{O}(Mm^2+\lceil\log Q\rceil) \tag{C10}$$

single and two-qubit gates.

Since we now have both block-encodings of $\widetilde{\mathcal{F}}(t_{j,q})$ and $\widetilde{\Lambda}(t_{j,q})$, we can implement $\texttt{HAM-T}_j$ with $\alpha_\Lambda = 5M/(2h_\xi^2)$, which is the $(\alpha_\Lambda, a_{\mathrm{HT}}, 4\alpha_\Lambda\epsilon')$-block-encoding where $a_{\mathrm{HT}} = \lceil\log(5M)\rceil + \lceil\log(Q)\rceil + 2a_{\mathrm{obj}} + 2\lceil\log\big((1/\nu)(e^{\nu S})/\epsilon'\big)\rceil + 4$ and requires

$$\tilde{\mathcal{O}}\left(\frac{1}{\nu}e^{\nu S}\alpha_{\mathrm{PDE}} d N^{\frac{\eta}{d}} M^2 T \log\left(\frac{e^{\nu S}}{\nu\epsilon'}\right)^{2+\frac{1}{\beta}} + Mm^2 + \lceil\log Q\rceil\right), \tag{C11}$$

single and two-qubit gates. The error $4\alpha_\Lambda\epsilon'$ comes from the product of block-encodings [42, Lemma 53]. By setting $\epsilon' \le \epsilon/(16\alpha_\Lambda N_{\mathrm{step}})$, the error of the block-encoding is bounded by $\epsilon/(4N_{\mathrm{step}})$, which satisfies the requirement for performing interaction picture simulation within error ϵ [35, Lemma 13].

Appendix D: Friction parameter setting by normalizing objective function

Quantum Hamiltonian descent employs the quantum counterpart of the classical dynamics governed by Newton's equation of motion

$$\ddot{\xi} + \nu\dot{\xi} + \nabla\mathcal{F} = 0, \tag{D1}$$

which describes the frictional dynamics driven by a force $\nabla\mathcal{F}$. Hence, the time to solution highly depends on the contrast of the friction parameter ν and the force $\nabla\mathcal{F}$, which makes it important to appropriately set the friction parameter based on the scale of $\mathcal{F}$. In this study, we set the friction parameter by normalizing Newton's equation of motion using the characteristic objective function value $\bar{\mathcal{F}}$. Dividing Newton's equation of motion by $\bar{\mathcal{F}}$, we obtain

$$\frac{1}{\bar{\mathcal{F}}}\frac{\mathrm{d}^2\xi}{\mathrm{d}t^2} + \frac{\nu}{\bar{\mathcal{F}}}\frac{\mathrm{d}\xi}{\mathrm{d}t} + \frac{1}{\bar{\mathcal{F}}}\nabla\mathcal{F} = 0. \tag{D2}$$

Then, we normalize time, the friction parameter, and the objective function as

$$\tilde{t} := \sqrt{\bar{\mathcal{F}}}t,\ \tilde{\nu} := \frac{\nu}{\sqrt{\bar{\mathcal{F}}}},\ \tilde{\mathcal{F}} := \frac{\mathcal{F}}{\bar{\mathcal{F}}}, \tag{D3}$$

which yields the normalized Newton's equation of motion

$$\frac{\mathrm{d}^2\xi}{\mathrm{d}\tilde{t}^2} + \tilde{\nu}\frac{\mathrm{d}\xi}{\mathrm{d}\tilde{t}} + \nabla\tilde{\mathcal{F}} = 0. \tag{D4}$$

Since this equation no longer depends on the scale of the objective function, we can set the friction parameter $\tilde{\nu}$ independent of the problem of interest. From empirical study, we employ $\tilde{\nu} = 1$, which therefore means that we set $\nu = \sqrt{\bar{\mathcal{F}}}$ for QHD. As the characteristic objective function value, we use the mean value of the objective function for all possible parameters, that is,

$$\begin{aligned}\bar{\mathcal{F}} &= \frac{1}{2^{Mm}}\sum_{\xi}\mathcal{F}(\xi)\\ &= \alpha_{\mathrm{obj}}\left(\langle\psi(0)|\langle 0|^{\otimes a_{\mathrm{obj}}} U_{\mathrm{obj}} |\psi(0)\rangle|0\rangle^{\otimes a_{\mathrm{obj}}} + 1\right),\end{aligned} \tag{D5}$$

where $|\psi(0)\rangle = 2^{-Mm/2}\sum_\xi |\xi\rangle$ is the initial state of QHD in this study. This value can be estimated by $\mathcal{O}(\varepsilon^{-1})$ queries to U_{obj} and $\mathcal{O}(\varepsilon^{-1})$ times state preparation to achieve precision ε [50].

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